

FANUC Robot series

R-30iB PLUS AND R-30iB MATE PLUS CONTROLLER

iRTorchMate

Setup and Operations Manual

MAROURTCH05181E REV A

This publication contains proprietary information of FANUC America Corporation furnished for customer use only. No other uses are authorized without the express written permission of FANUC America Corporation.

FANUC America Corporation
3900 W. Hamlin Road
Rochester Hills, Michigan 48309-3253

Copyrights and Trademarks

This new publication contains proprietary information of FANUC America Corporation furnished for customer use only. No other uses are authorized without the express written permission of FANUC America Corporation.

The descriptions and specifications contained in this manual were in effect at the time this manual was approved for printing. FANUC America Corporation, hereinafter referred to as FANUC, reserves the right to discontinue models at any time or to change specifications or design without notice and without incurring obligations.

FANUC manuals present descriptions, specifications, drawings, schematics, bills of material, parts, connections and/or procedures for installing, disassembling, connecting, operating and programming FANUC products and/or systems. Such systems consist of robots, extended axes, robot controllers, application software, the KAREL® programming language, INSIGHT® vision equipment, and special tools.

FANUC recommends that only persons who have been trained in one or more approved FANUC Training Course(s) be permitted to install, operate, use, perform procedures on, repair, and/or maintain FANUC products and/or systems and their respective components. Approved training necessitates that the courses selected be relevant to the type of system installed and application performed at the customer site.



WARNING

This equipment generates, uses, and can radiate radiofrequency energy and if not installed and used in accordance with the instruction manual, may cause interference to radio communications. As temporarily permitted by regulation, it has not been tested for compliance with the limits for Class A computing devices pursuant to Subpart J and Part 15 of FCC Rules, which are designed to provide reasonable protection against such interference. Operation of the equipment in a residential area is likely to cause interference, in which case the user, at his own expense, will be required to take whatever measure may be required to correct the interference.

FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

FANUC America Corporation
Training Department
3900 W. Hamlin Road
Rochester Hills, Michigan 48309-3253
www.fanucamerica.com

For customer assistance, including Technical Support, Service, Parts & Part Repair, and Marketing Requests, contact the Customer Resource Center, 24 hours a day, at 888-FANUC-US (888-326-8287).

Send your comments and suggestions about this manual to:
product.documentation@fanucamerica.com

**Copyright © 2018 by FANUC America Corporation
All Rights Reserved**

The information illustrated or contained herein is not to be reproduced, copied, downloaded, translated into another language, published in any physical or electronic format, including internet, or transmitted in whole or in part in any way without the prior written consent of FANUC America Corporation.

AccuStat®, ArcTool®, iRVision®, KAREL®, PaintTool®, PalletTool®, SOCKETS®, SpotTool®, SpotWorks®, and TorchMate® are Registered Trademarks of FANUC.

FANUC reserves all proprietary rights, including but not limited to trademark and trade name rights, in the following names:

AccuAir™, AccuCal™, AccuChop™, AccuFlow™, AccuPath™, AccuSeal™, ARC Mate™, ARC Mate Sr.™, ARC Mate System 1™, ARC Mate System 2™, ARC Mate System 3™, ARC Mate System 4™, ARC Mate System 5™, ARCWorks Pro™, AssistTool™, AutoNormal™, AutoTCP™, BellTool™, BODYWorks™, Cal Mate™, Cell Finder™, Center Finder™, Clean Wall™, DualARM™, iRProgrammer™, LR Tool™, MIG Eye™, MotionParts™, MultiARM™, NoBots™, Paint Stick™, PaintPro™, PaintTool 100™, PAINTWorks™, PAINTWorks II™, PAINTWorks III™, PalletMate™, PalletMate PC™, PalletTool PC™, PayloadID™, RecipTool™, RemovalTool™, Robo Chop™, Robo Spray™, S-420i™, S-430i™, ShapeGen™, SoftFloat™, SOFT PARTS™, SpotTool+™, SR Mate™, SR ShotTool™, SureWeld™, SYSTEM R-J2 Controller™, SYSTEM R-J3 Controller™, SYSTEM R-J3iB Controller™, SYSTEM R-J3iC Controller™, SYSTEM R-30iA Controller™, SYSTEM R-30iA Mate Controller™, SYSTEM R-30iB Controller™, SYSTEM R-30iB Mate Controller™, SYSTEM R-30iB Plus Controller™, SYSTEM R-30iB Mate Plus Controller™, TCP Mate™, TorchMate™, TripleARM™, TurboMove™, visLOC™, visPRO-3D™, visTRAC™, WebServer™, WebTP™, and YagTool™.

©FANUC CORPORATION 2018

- No part of this manual may be reproduced in any form.
- All specifications and designs are subject to change without notice.

Patents

One or more of the following U.S. patents might be related to the FANUC products described in this manual.

FANUC America Corporation Patent List

4,630,567 4,639,878 4,707,647 4,708,175 4,708,580 4,942,539 4,984,745
5,238,029 5,239,739 5,272,805 5,293,107 5,293,911 5,331,264 5,367,944
5,373,221 5,421,218 5,434,489 5,644,898 5,670,202 5,696,687 5,737,218
5,823,389 5,853,027 5,887,800 5,941,679 5,959,425 5,987,726 6,059,092
6,064,168 6,070,109 6,086,294 6,122,062 6,147,323 6,204,620 6,243,621
6,253,799 6,285,920 6,313,595 6,325,302 6,345,818 6,356,807 6,360,143
6,378,190 6,385,508 6,425,177 6,477,913 6,490,369 6,518,980 6,540,104
6,541,757 6,560,513 6,569,258 6,612,449 6,703,079 6,705,361 6,726,773
6,768,078 6,845,295 6,945,483 7,149,606 7,149,606 7,211,978 7,266,422
7,399,363

FANUC CORPORATION Patent List

4,571,694 4,626,756 4,700,118 4,706,001 4,728,872 4,732,526 4,742,207
4,835,362 4,894,596 4,899,095 4,920,248 4,931,617 4,934,504 4,956,594
4,967,125 4,969,109 4,970,370 4,970,448 4,979,127 5,004,968 5,006,035
5,008,834 5,063,281 5,066,847 5,066,902 5,093,552 5,107,716 5,111,019
5,130,515 5,136,223 5,151,608 5,170,109 5,189,351 5,267,483 5,274,360
5,292,066 5,300,868 5,304,906 5,313,563 5,319,443 5,325,467 5,327,057
5,329,469 5,333,242 5,337,148 5,371,452 5,375,480 5,418,441 5,432,316
5,440,213 5,442,155 5,444,612 5,449,875 5,451,850 5,461,478 5,463,297
5,467,003 5,471,312 5,479,078 5,485,389 5,485,552 5,486,679 5,489,758
5,493,192 5,504,766 5,511,007 5,520,062 5,528,013 5,532,924 5,548,194
5,552,687 5,558,196 5,561,742 5,570,187 5,570,190 5,572,103 5,581,167
5,582,750 5,587,635 5,600,759 5,608,299 5,608,618 5,624,588 5,630,955
5,637,969 5,639,204 5,641,415 5,650,078 5,658,121 5,668,628 5,687,295
5,691,615 5,698,121 5,708,342 5,715,375 5,719,479 5,727,132 5,742,138
5,742,144 5,748,854 5,749,058 5,760,560 5,773,950 5,783,922 5,799,135
5,812,408 5,841,257 5,845,053 5,872,894 5,887,122 5,911,892 5,912,540
5,920,678 5,937,143 5,980,082 5,983,744 5,987,591 5,988,850 6,023,044
6,032,086 6,040,554 6,059,169 6,088,628 6,097,169 6,114,824 6,124,693
6,140,788 6,141,863 6,157,155 6,160,324 6,163,124 6,177,650 6,180,898
6,181,096 6,188,194 6,208,105 6,212,444 6,219,583 6,226,181 6,236,011
6,236,896 6,250,174 6,278,902 6,279,413 6,285,921 6,298,283 6,321,139
6,324,443 6,328,523 6,330,493 6,340,875 6,356,671 6,377,869 6,382,012
6,384,371 6,396,030 6,414,711 6,424,883 6,431,018 6,434,448 6,445,979
6,459,958 6,463,358 6,484,067 6,486,629 6,507,165 6,654,666 6,665,588
6,680,461 6,696,810 6,728,417 6,763,284 6,772,493 6,845,296 6,853,881
6,888,089 6,898,486 6,917,837 6,928,337 6,965,091 6,970,802 7,038,165
7,069,808 7,084,900 7,092,791 7,133,747 7,143,100 7,149,602 7,131,848
7,161,321 7,171,041 7,174,234 7,173,213 7,177,722 7,177,439 7,181,294
7,181,313 7,280,687 7,283,661 7,291,806 7,299,713 7,315,650 7,324,873
7,328,083 7,330,777 7,333,879 7,355,725 7,359,817 7,373,220 7,376,488
7,386,367 7,464,623 7,447,615 7,445,260 7,474,939 7,486,816 7,495,192
7,501,778 7,502,504 7,508,155 7,512,459 7,525,273 7,526,121

Conventions



WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.



CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
- Using a low motion speed, test run the program continuously for at least one full cycle.
- Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

Table of Contents

1. iRVISION TORCHMATE APPLICATION OVERVIEW	1-1
1.1. OVERVIEW.....	1-1
1.2. HOW iRVISION TORCH MATE WORKS	1-1
1.2.1. Single Wire Torch.....	1-1
1.2.2. Dual Wire torch	1-2
1.3. INSPECTION BOX	1-3
2. iRVISION TORCHMATE PRODUCT FAMILY.....	2-1
2.1. iRVISION.....	2-1
3. FEATURES.....	3-1
4. KEY CONCEPTS AND DEFINITIONS	4-1
4.1. OVERVIEW.....	4-1
4.2. WORKCELL.....	4-1
4.3. VISION EDGE POINTS TOOL	4-1
4.4. MODEL NUMBERS	4-1
4.5. ROBOT	4-2
4.6. INSPECTION STATION	4-2
4.7. USER FRAMES	4-2
4.7.1. APP Frame.....	4-2
4.7.2. TEMP Frame.....	4-2
4.8. USER TOOLS.....	4-3
4.8.1. VIEW Tool	4-3
4.8.2. GRID Tool	4-3
4.8.3. SINGLE Tool.....	4-3
4.8.4. MASTER Single Tool	4-4
4.8.5. DUAL1 and DUAL2 Tools	4-4
4.8.6. MASTER Dual1 and MASTER Dual2 Tools.....	4-4
4.8.7. Nominal Tools	4-4
5. HARDWARE SETUP.....	5-1
5.1. OVERVIEW.....	5-1
5.2. INSPECTION STATION SETUP (STEP 1)	5-1

5.3.	CONNECT CAMERA CABLE (STEP 2)	5-2
5.4.	CONNECT LIGHT POWER CABLE (STEP 3).....	5-2
5.5.	PADDLE GRID	5-2
6.	SOFTWARE SETUP	6-1
6.1.	OVERVIEW.....	6-1
6.2.	INSTALL IRVISION TORCHMATE SOFTWARE.....	6-1
6.3.	RUN IRTSETUP_T1	6-1
6.4.	SET UP IRVISION TORCHMATE DATABASE	6-2
6.4.1.	Enter the database	6-2
6.4.2.	Navigating the database	6-3
6.4.3.	Default Setup	6-4
6.4.4.	Modifying Lamp Control.....	6-5
6.5.	SETUP VISION DATA	6-6
6.5.1.	Setup Camera Data	6-6
6.5.2.	Attach the grid paddle to the torch.....	6-8
6.5.3.	Teach Grid Tool as the Calibration Grid Tool.....	6-9
6.5.4.	Teach the User Frame of the Inspection box	6-11
6.5.5.	Perform Camera Calibration	6-13
6.5.6.	Teach Master UTOOL(S)	6-17
6.5.7.	Set up Vision Process	6-17
7.	VISION	7-1
7.1.	OVERVIEW.....	7-1
7.2.	CAMERA DATA.....	7-1
7.3.	VISION PROCESS SETUP	7-1
7.3.1.	Train the Vision Process	7-1
7.3.2.	Setup the Snap Tool.....	7-3
7.3.3.	Train the Selected Edge Points Tool.....	7-4
7.3.4.	Set up the Selected Edge Points Tool	7-4
7.3.5.	Set the Search Window.....	7-4
7.3.6.	Test the Vision Process.....	7-5
8.	GRAPHIC USER INTERFACE.....	8-1
8.1.	OVERVIEW.....	8-1

8.1.1.	Setup Menu	8-1
8.1.2.	Status Menu	8-2
8.2.	IRVISION TORCHMATE THRESHOLD SCREEN	8-2
8.3.	IRVISION TORCHMATE TORCH SETUP SCREEN.....	8-3
8.4.	IRVISION TORCHMATE WIRE SETUP SCREEN	8-4
8.5.	IRVISION TORCHMATE VIEW SETUP SCREEN	8-6
8.6.	(CONTROL) FLAGS SETUP	8-7
8.7.	STATUS SCREEN	8-9
8.8.	MODEL STATUS SCREEN	8-11
8.9.	CURRENT TOOL SCREEN	8-12
8.10.	NOMINAL TOOL SCREEN	8-12
8.11.	TOOL OFFSET SCREEN.....	8-13
8.12.	INVERSE TOOL OFFSET SCREEN.....	8-14
9.	SAVING AND RESTORING	9-1
9.1.	TP PROGRAMS.....	9-1
9.2.	KAREL DATA.....	9-1
9.3.	VISION DATA.....	9-1
10.	LOGGING	10-1
10.1.	OVERVIEW.....	10-1
10.2.	ACTIVATING THE LOGGING.....	10-1
10.3.	FILE MAINTENANCE	10-1
10.4.	IRVISION TORCHMATE.CSV LOG DATA	10-1
10.5.	RESETTING THE LOG COUNT.....	10-2
A	ERROR CODES.....	A-1
B	PROGRAMMING AND CUSTOMIZING IRVISION TORCHMATE INSPECTION.....	B-1
B.1	CALLING A FUNCTION	B-1
B.1.1	KAREL Call	B-1
B.1.2	TP Call	B-1
B.2	FUNCTION DESCRIPTION AND ARGUMENT LIST	B-2
B.3	IRTMAIN FUNCTIONS	B-3
B.3.1	Inquire	B-3
B.3.2	Logging.....	B-3

B.3.3	RunSnap1.....	B-4
B.3.4	RunSnap2.....	B-4
B.3.5	GetTool	B-4
B.3.6	Init.....	B-5
B.3.7	SetTorch.....	B-5
B.3.8	SetVPName.....	B-5
B.3.9	SetGroup	B-6
B.3.10	SetNom	B-6
C	USING IRTSETUP_T1	C-1
C.2	SET TORCH NUMBER.....	C-1
C.3	STEP 1. INSTALL INSPECTION BOX.....	C-1
C.4	STEP 2. CONNECT CAMERA CABLE.....	C-2
C.5	STEP 3. CONNECT LED LIGHT CABLE	C-2
C.6	INITIALIZE ALL KAREL DATA	C-2
C.7	STEP 4. SETUP IRTORCHMATE PARAMETERS	C-2
C.8	STEP 5. TRAIN THE MASTER UTOOLS	C-3
C.9	STEP 6. RELOCATE THE CAMERA IN IRVISION.....	C-3
C.10	STEP 7. SETUP CAMERA (IRTCALIB_T1).....	C-3
C.11	STEP 8. INSTALL CALIBRATION GRID	C-4
C.12	STEP 9. TEACH UTOOL[7].....	C-4
C.13	STEP 10. REMOVE THE CAMERA	C-4
C.14	STEP 11. CHECK AND RE-ADJUST EXPOSURE	C-4
C.15	STEP 12. TEACH UFRAME OF IRVISION BOX.....	C-5
C.16	STEP 12.1 IRTSETFRM.TP	C-5
C.17	STEP 13 SETUP IRTCALIB_T1 IN VISION	C-6
C.18	STEP 13.1 IRTCALIB_T1.TP	C-6
C.19	STEP 14. REMOVE GRID FROM TORCH NECK	C-7
C.20	STEP 15. TRAIN UTOOL[4] USING SIX POINT METHOD	C-7
C.21	TRAIN UTOOL[1 & 2] USING SIX POINT METHOD IN CASE OF A TANDEM TCP	C-7
C.22	STEP 16. TRAIN VISION PROCESS.....	C-8
C.23	STEP 17. GET NUMBER OF WIRES.....	C-9
C.24	STEP 18. SET MASTER TOOL TO SINGLE UTOOL.....	C-9
C.25	STEP 18.1 IRTUTILITY_T1(140).....	C-9

C.26	MANUALLY OVERRIDE MASTER UTOOL	C-10
C.27	STEP 19. SET THE MASTER UTOOLS[9 & 10] TO THE DUAL UTOOLS[1 & 2]	C-11
C.28	STEP 20. MANUALLY OVERRIDE THE MASTER UTOOLS	C-11
C.29	SET NOMINAL FLAG.....	C-11
C.30	IRT_T1.TP.....	C-11
C.31	SETUP COMPLETE.....	C-13
D	USING IRTSETUP_G1	D-1
D.2	INTRODUCTION.....	D-1
D.3	INSTALL HARDWARE	D-1
D.4	INITIALIZING THE DATABASE.....	D-1
D.5	SETUP THE DATABASE	D-2
D.6	TRAIN CALIBRATION GRID	D-2
D.7	TRAIN THE APPLICATION FRAME.....	D-3
D.8	CAMERA CALIBRATION USING IRTCALIB_G1.....	D-4
D.9	REMOVE CALIBRATION GRID PADDLE	D-5
D.10	TRAIN USER TOOLS.....	D-5
D.10.11	Train Master Single Wire User Tool	D-5
D.10.12	Train View User Tool (for single wire).....	D-6
D.10.13	Train View User Tool for Tandem Torch.....	D-6
D.10.14	Train Master Dual Wire 1 User Tool.....	D-7
D.10.15	Train Master Dual Wire 2 User Tool.....	D-7
D.11	TRAIN VISION PROCESS	D-7
D.12	UPDATE THE MASTER TOOLS AS GOLDEN TOOLS	D-11
D.13	SET THE NOMINAL TOOL (GOLDEN WIRE TOOLS)	D-13

1

IRVISION TORCHMATE APPLICATION OVERVIEW

1. IRVISION TORCHMATE APPLICATION OVERVIEW

1.1. Overview

iRVision TorchMate (R744) is designed to inspect and locate the weld torch wires visually from two camera views. The torch may be a single wire torch or a dual wire torch.

The two camera views provide a front and side view that triangulates the location of the wires tip. Instead of having two cameras, the robot is rotated 90 degrees to provide the same information as a front and side camera views. The front and side views are actually -45 and +45 degrees from the front view. This saves the cost of an additional camera and requires less floor space.

1.2. How iRVision Torch Mate Works

The robot is moved near the Inspection box opening. The iRVision TorchMate program moves the torch into the inspection box to the first view position. A camera image is acquired and the robot moves to a second view position. The two view positions are 90 degrees apart. A second image is acquired.

The two views are triangulated to define the locations of both wires. The UTOOLS for both wires are updated if successful. In some cases a retest is automatically tried. If an error results, the robot exits the box and waits for operator intervention.

1.2.1. Single Wire Torch

For a single torch wire, a set of points is generated along the outside of the wire. A best fit line is calculated for all the points. The lowest point is projected onto the line. This defines the UTOOL origin. The X axis direction defaults to the X direction of the face plate. The UTOOL can be relocated a fixed X Z distance or rotated a fixed Roll angle about the wire.

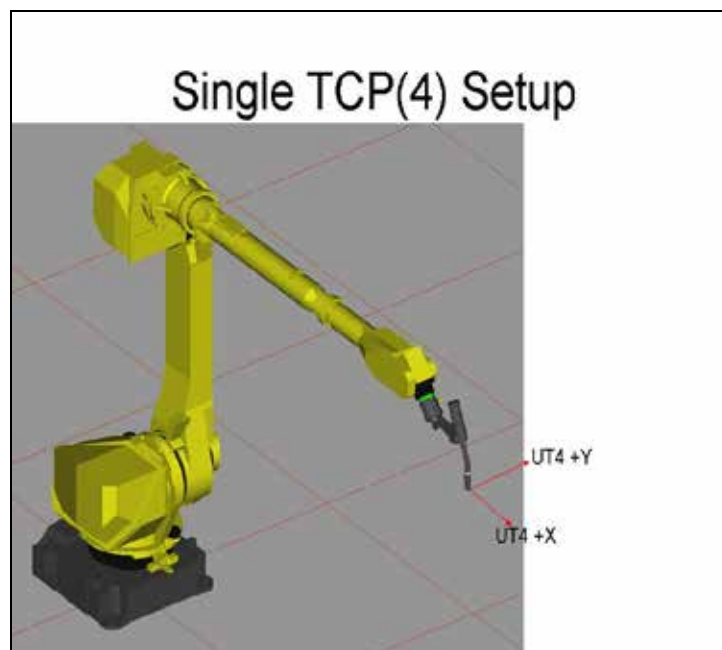


Figure 1-1 Single Wire UTOOL Vectors

1.2.2. Dual Wire torch

For a dual wire torch, a set of points is generated along the outside of both wires. A best fit line is calculated for all the points for each wire. The lowest point for each wire is projected onto its line. This defines the two UTOOL origins.

The X axis direction is defined to pass through both wire tips. The UTOOL can be relocated a fixed X or Z distance or rotated a fixed Roll angle.

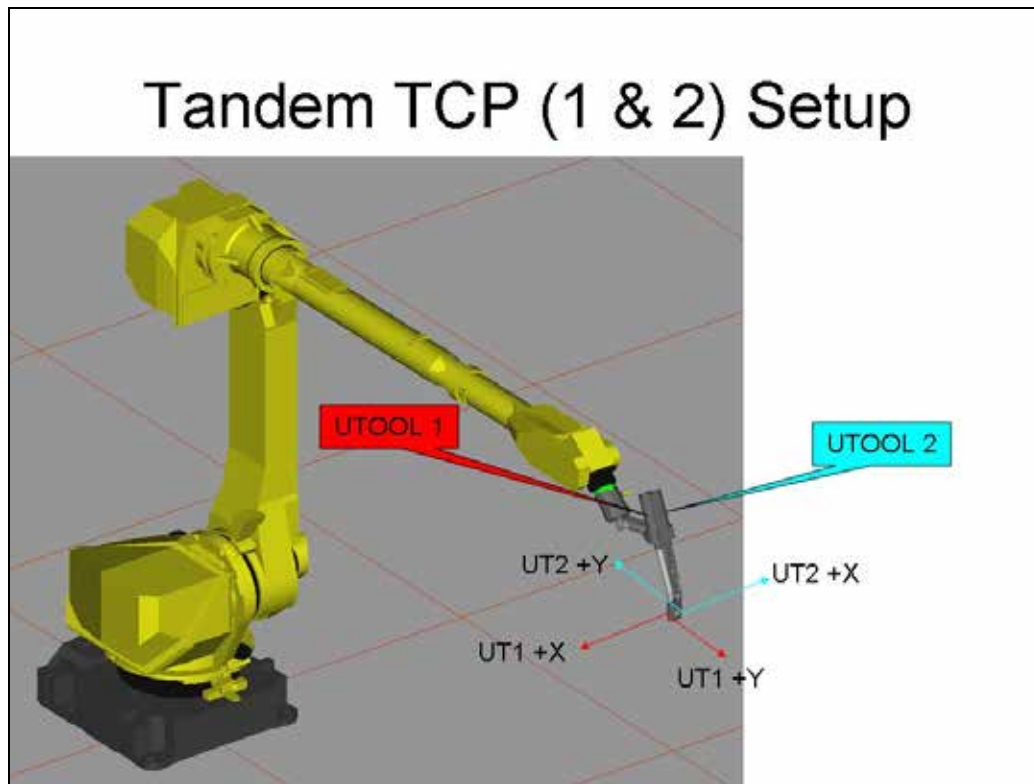


Figure 1-2 Dual Wire Torch Vectors

1.3. Inspection Box

The inspection box provides an ideal environment for lighting and imaging the torch wires while protecting them from the environment and shading the wires from ambient light. Instead of using two cameras at two angles, the same image information can be obtained by rotating the torch inside the inspection box. This saves the cost of an extra camera, and saves floor space.

Figure 1-3 shows the application frame that has been taught relative to the inspection box. The magenta and cyan lines represent the two view positions of the torch.

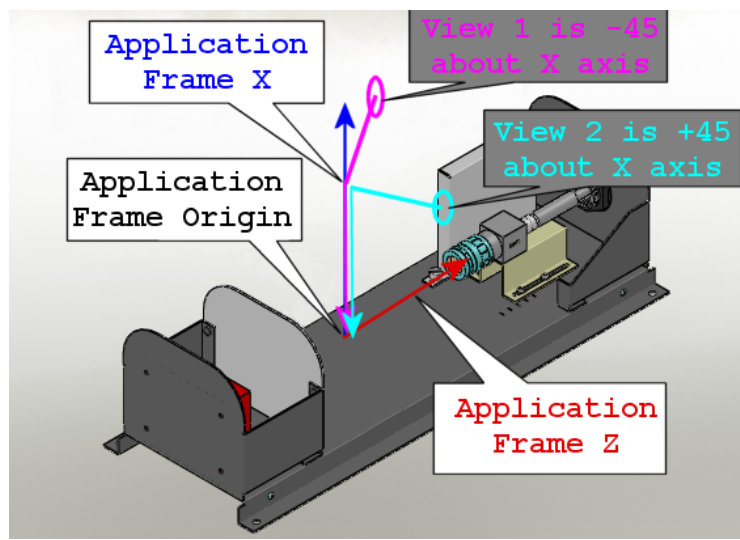


Figure 1-3 Inspection Box (top removed)

2

IRVISION TORCHMATE PRODUCT FAMILY

2. IRVISION TORCHMATE PRODUCT FAMILY

2.1. IRVision

IRVision is a visual sensor system designed to provide robot guidance and error proofing capabilities by inspecting a work piece using cameras. It can perform various measurements as well as adjust the robot motion so that the robot can manipulate the work piece in the same way as programmed. For full details about the setup and use of *iRVision*, please refer to the *iRVision Operator* manuals.

3

FEATURES

3. FEATURES

- Fully integrated *iR*Vision product
- Supports four Single and/or Dual wire torches
- Requires less than five seconds to update tandem or single torch UTOOLS (includes robot motion time in the box)
- Camera and lens are enclosed in the Inspection box allowing for ideal lighting and imaging of the torch wires while offering protection from the environment.
- Warnings and/or errors are logged when the UTOOL elements (XYZWPR) vary too far from an ideal master tool.
- Supports torch exchange

4

KEY CONCEPTS AND DEFINITIONS

4. KEY CONCEPTS AND DEFINITIONS

4.1. Overview

The iRVision TorchMate product is designed to view the torch wire visually to determine its location and orientation relative to the face plate of the robot. The UTOOL of the torch wire is updated. If the torch has two wires (tandem), then two UTOOLS are updated.

4.2. Workcell

The workcell consists of a robot with a weld torch. The robot performs a number of welds. After welding the robot moves to a clip station and the wires are cut to a predefined length.

The spooling of the wire can cause the wire to relocate as wire is fed through the torch. The end position may vary. The robot places the weld torch into the Inspection Box. The new location for the UTOOL is determined.

4.3. Vision Edge Points Tool

The Vision Edge Points tool is a unique vision tool used to find all the pixels that contain a wire and white background. Only the pixels in an area of interest are analyzed. This area is a generous sized rectangle that includes the wires and background. The torch proper should be excluded from the search area. The wires are illuminated with back lighting, allowing a silhouette of the wires to be acquired. The wire for tandem operations on the right side of the Field of View (FOV) is wire #1. The second wire is wire #2. The wire numbers can be assigned to any UTOOL number.

4.4. Model Numbers

Four models are defined for the two wires. Model #1 is always the bottom of wire #1. Model #3 is always the highest point on Wire #1 in the area of interest. Wire #1 is the right hand wire for Dual torches or the single wire torch. Model #4 is the lowest point on Wire #2. Model #6 is the highest point on Wire #2 in the area of interest. Wire #2 is the left hand wire of a dual torch.

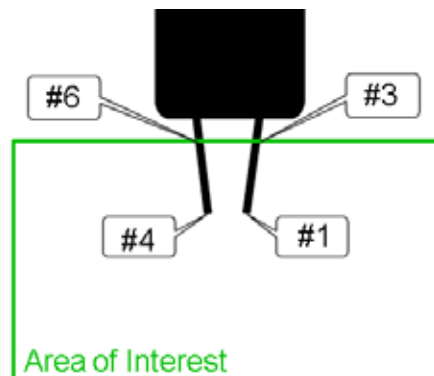


Figure 4-1 Wire Model Numbers

4.5. Robot

The robot is responsible for delivering the torch to the correct position. Robot software allows it to measure the positions of the wires and to modify the UTOOL to ensure proper welding start position. The robot can be programmed to periodically inspect the weld wires and adjust accordingly by moving near the Inspection Station and calling the IRT_T1.TP program.

4.6. Inspection Station

The weld wire measurements are performed in the inspection station. The station consists of a camera and light source in an enclosure. The robot moves to the *Enter/Exit* position near the Inspection station to ensure a direct path into the box. The robot moves linearly to *View 1* position inside the camera and light enclosure, an image is acquired. The robot moves linearly to *View 2*, inside the camera and light enclosure. A second image is acquired. The robot moves to the *Enter/Exit* position. The user tools are calculated.

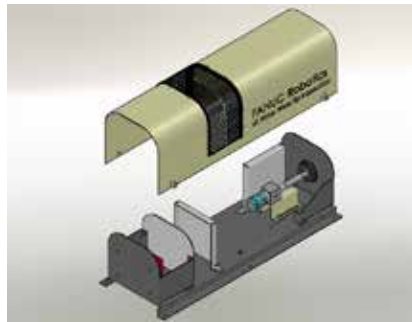


Figure 4-2 Inspection Station

4.7. User Frames

Two user frames are used by the iRVision Torch Mate software. The database has the frame number (\$MNUFRAMENUM) assigned to a string name. The string name is used throughout the system by routines for changing the frame number. The header sections below are named after their string names.

4.7.1. APP Frame

The application frame or Inspection box frame is taught to the Inspection Box. By default UFRAME [1] is used. This value can be changed by defining the Application frame number in the SETUP FLAGS screen. The Z axis is toward the camera and aligned with the axis of the camera lens. The X axis is upward. The origin is in the center of the opening.

4.7.2. TEMP Frame

The TEMP frame or temporary is taught to teach UTOOL in the Inspection Box. By default UFRAME [5] is used. This value can be changed by defining the TEMP frame number in the SETUP FLAGS screen. It is a throw away frame and is not used for robot moves. It is used for calculating user tools from the current position.

4.8. User Tools

Eight user tools are used by the IRVision Torch Mate software. The database has the tool number (\$MNUTOLNUM) assigned to a string name. The string name is used throughout the system by routines for changing the tool number. The header section below is named after their string names. By reassigning the Tool number to the string name the user can avoid editing the code to change user tool numbers.

4.8.1. VIEW Tool

By default the VIEW Tool is assigned to UTOOL[3]. The value can be changed by defining the View Tool number in the SETUP FLAGS screen. The tool is used when moving the robot to the above inspection box position and the two view positions in the box. The tool's XYZ is updated before entering the inspection box. If a Single wire torch is in use, then the XYZ matches the XYZ of the Torch Tool. If a dual wire torch is in use, then the XYZ is the mid-point of the two wire tips. The WPR is not updated. When it is originally taught during SETUP, the Tool's orientation aligned with the Application frames orientation. That is, when the robot is at the over box position, jogging in UTOOL [3] or Application Frame (UFRAME[1]) jogs in the same XY and Z direction. This allows the same position (UTOOL and UFRAME) to be used for all torches in the same program. The over Box position is usually (100, 0, 0, 0, 0, 0). The View 1 position is (0, 0, 0, -45, 0, 0) and the View 2 position is (0, 0, 0, 45, 0, 0)

4.8.2. GRID Tool

The Grid tool defines the location of the calibration grid that is mounted on a Plexiglas paddle (see Figure 4-3 Grid Tool). This tool is used to calibrate the camera and define the application frame. If the user changes this value be sure to edit the *Tool Frame* value CAMERA CALIBRATION SETUP (IRTCALIB_T1.VD) vision data file to match the new tool number. The *Camera User Tool Number* in the AUTOMATIC GRID FRAME SET utility, and the Setup> Flags> screen must also be changed.

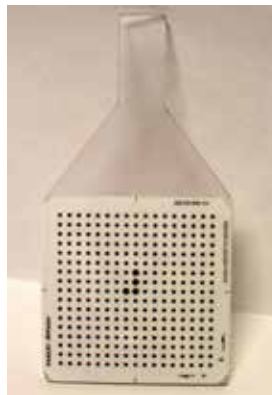


Figure 4-3 Grid Tool

4.8.3. SINGLE Tool

The SINGLE tool is assigned to UTOOL[4] by default. It is the UTOOL used when teaching a weld path with a single wire torch. It can be assigned to a different UTOOL Number by using the WIRE SETUP screen. This UTOOL is updated whenever iRVision TorchMate is executed on a single wire torch.

4.8.4. MASTER Single Tool

This tool is referred to as MASTERS. It is used to override elements of the Single tool after a calculation. By default it is assigned to UTOOL [8]. It can be assigned to a different UTOOL Number by using the WIRE SETUP screen.

4.8.5. DUAL1 and DUAL2 Tools

The DUAL1 tool is assigned to UTOOL[1] by default. It is the UTOOL used when teaching a weld path with the first wire of a dual wire torch. It can be assigned to a different UTOOL Number by using the WIRE SETUP screen. This UTOOL is updated whenever iRVision TorchMate is executed on a dual wire torch. By convention this is the wire on the right side of the camera image.

The DUAL2 tool is assigned to UTOOL[2] by default. It is the UTOOL used when teaching a weld path with the second wire of a dual wire torch. It can be assigned to a different UTOOL Number by using the WIRE SETUP screen. This UTOOL is updated whenever iRVision TorchMate is executed on a dual wire torch. By convention this is the wire on the left side of the camera image.

The two tools' X axis are 180 degrees apart. If the user wants the right side wire to be UTOOL[2] and the left side wire to be UTOOL[1] switch the assignments in the WIRE SETUP screen.

4.8.6. MASTER Dual1 and MASTER Dual2 Tools

This tool is referred to as MASTERS. It is used to override elements of the Single tool after a calculation. By default they are assigned to UTOOL [9] and UTOOL[10]. It can be assigned to a different UTOOL Number by using the WIRE SETUP screen.

4.8.7. Nominal Tools

TorchMate will calculate the UTOOL as long as the wire tips appear in the search window. When the tool is first measured, a copy of its value is stored for later comparisons. This first Utool value is the nominal tool. All subsequent calculations will compare the results of the measured tool with the nominal tool. This will determine how much the Tool is drifting over time and may indicate a damaged tool.

5

HARDWARE SETUP

5. HARDWARE SETUP

5.1. Overview

Hardware setup has been condensed to three general steps.

- Place the inspection station at a convenient location.

- Connect Camera Cable.

- Connect Light Power cable.

- To prepare for calibration, the grid paddle needs to be attached to the torch.

- After calibration has been completed, the grid paddle is removed.

5.2. Inspection Station Setup (step 1)

The Inspection Station Box (see Figure 5-1) can be mounted anywhere in the cell. The position should be selected so that the robot can rotate the tool +45 degrees and -45 degrees while in the box. Figure 5-1 shows the inspection station box and detachable cover. The robot positions the torch in the box by passing through the brushes. The brushes help keep ambient light and debris out.

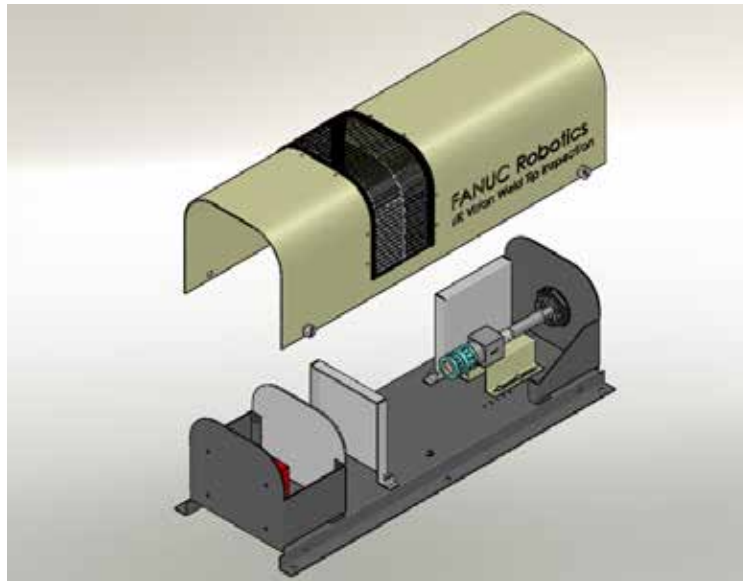


Figure 5-1 Inspection Station Box

The station should be placed so that minimum robot travel time is required. Normally the measurement operation can be preformed at intervals following the completion of all welds for the part.

The user should use the provide template program, *IRT.TP* or make a copy of the program. Programming a robot for performing a measurement consists of the following.

1. Move robot to the inspection *entrance / exit* position
2. Move robot linearly into the station's view position 1
3. Camera acquires image 1
4. Move robot linearly into the station's view position 2
5. Camera acquires image 2
6. Robot moves to the *entrance / exit* position
7. Robot performs recovery or continues weld operations.

5.3. Connect Camera Cable (step 2)

Once the inspection station is in position, the hardware can be connected. The controller to camera cable connectors are clearly labeled on the cable. Plug in the camera end of the cable to the camera, and the controller end to the coax BNC female connector on the robot controller CPU. For more information consult *Connecting a Camera* in the *Setup* section of the *iRVision Operator's Manual*. Please turn off the power to the controller when connecting and disconnecting the camera.

5.4. Connect Light Power Cable (step 3)

The light in the inspection station requires a 24VDC source. The user must connect the light power cable to a 24VDC source. A relay may be used to allow control of the light from an I/O device.

Hardware setup for the inspection box is complete.

5.5. Paddle Grid

Shown below is the paddle grid (Figure 5-2) that is attached to the torch during the calibration. It is attached with a simple hose clamp.

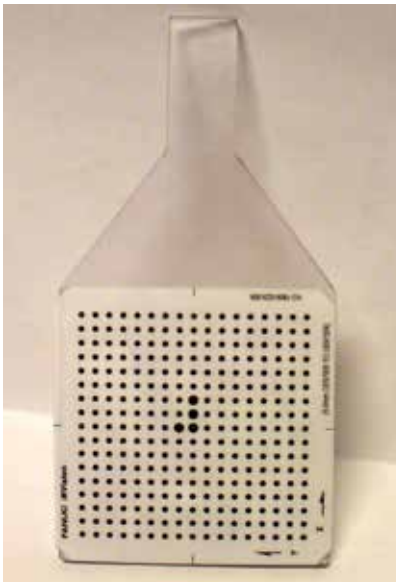


Figure 5-2 Calibration Grid Paddle

6

SOFTWARE SETUP

6. SOFTWARE SETUP

6.1. Overview

To complete the software setup successfully, several items must first need to be completed.

- Install Software
- Complete hardware setup (see previous chapter)
- Teach user frames and tool frames
- Teach Vision Data
- Set up *iRVision* Torch Mate Database
- Record robot positions.
- Set Reference Positions

6.2. Install *iRVision* TorchMate Software

Required Robot Option

- *iRVision* TorchMate Software (Robot option R744).

The *iRVision* TorchMate option requires either the *iRVision* 2D package or the *iRVision* 3D Package. The *iRVision* TorchMate option includes the KAREL programs (invisible to the user), TP templates of programs for performing the tests, Vision Data files, and the HTM screens. All of the files are loaded and initialized with default data when installed. System variables are set to link the Teach Pendant [Status] key to the HTM *Status* screen.

6.3. Run IRTSETUP_T1

To simplify the setup and complete all of the steps mentioned in 6.1, RUN the IRTSETUP.TP program in step mode. IRTSETUP_T1.TP is a template program used to Setup the *iRVision* Torch Mate. It provides a step by step set of instructions/comments indicating when to setup the database items or train vision data. In some cases the program calls other TP and KAREL programs to perform teaching of frames and tools, and vision calibration. The first part of the program should be executed to setup the cell. This is only performed once. The second part of the program is run for every torch setup.

6.4. Set up IRVision TorchMate Database

The settings in the database are used to

- Define the user frame numbers and the tool numbers.
- Describe tool orientation and origin of a wire.
- Define what elements of the master tool override the wire tool
- Define warning thresholds, error thresholds, and DOUT annunciators.
- Describe torch layout (number of wires and group number)
- Define vision process associated with Torch.

What follows is a quick guide to set up the data base

6.4.1. Enter the database

To gain access to the database, select the [STATUS] key of the Teach Pendant (See Figure 6-1).



Figure 6-1 Status Key – enclosed in orange box

One of the status screens will display.

If the AXIS status screen displays, press the [F1:TYPE] key and select the iRVision TM item. The IRVISION Torch Mate status screen will be displayed.

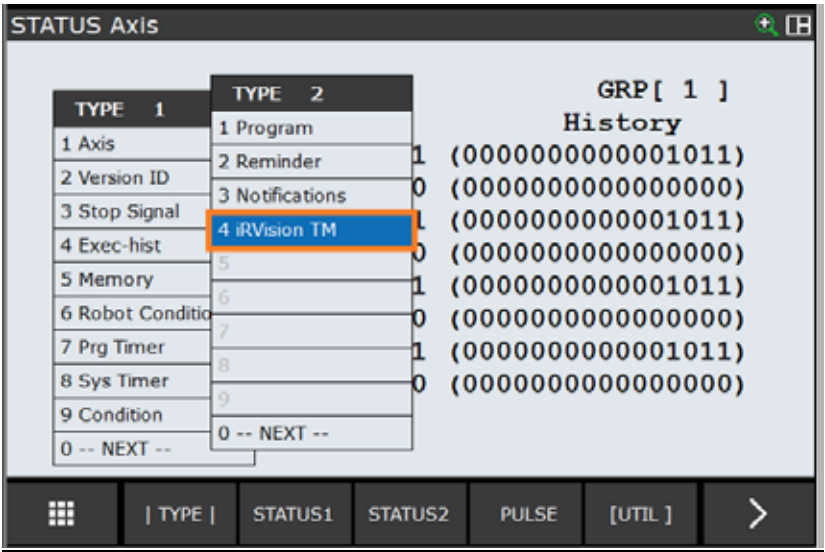


Figure 6-2 Status Screen TYPE selection

The Status screen for iRVision TorchMate is shown in Figure 6-3. This screen is the main entry point for all iRVision TorchMate screens. Any iRVision Torch Mate screen can be reached by using the two Navigation buttons at the bottom of the bottom of the display.

6.4.2. Navigating the database

The two navigation buttons are SETUP and STATUS. They are shown in Figure 6-3. SETUP allows the user to select one of the SETUP screens. STATUS allows the user to select one of the RUNTIME screens. Only the SETUP screens have fields that allow editing.

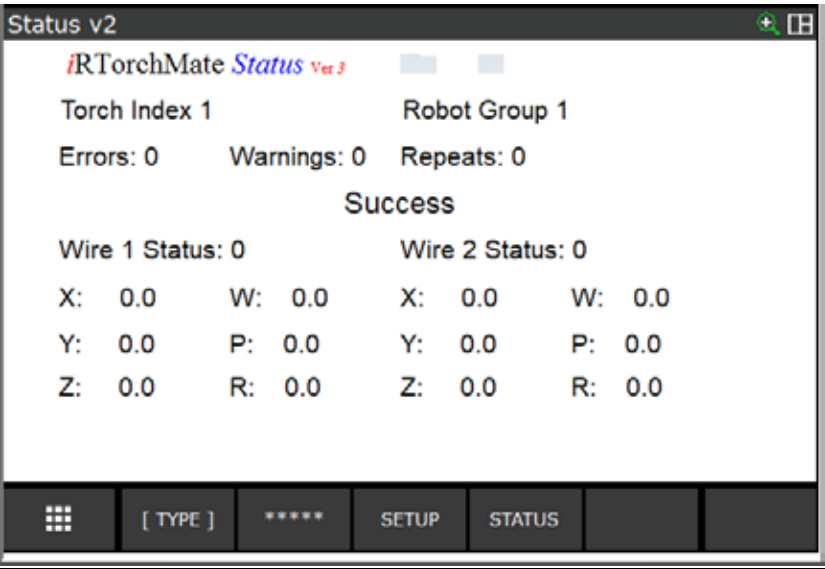


Figure 6-3 Status Screen for iRVision Torch Mate

6.4.3. Default Setup

The default setup assumes that the torch has one wire. The UTOOL will keep the same orientation as the master UTOOL and only the origin (tip of wire) is updated by the iRVision TorchMate test.

The default UTOOL is 1 and 2 for a Dual wire and 4 for a single wire. The UFRAME number is 1. Use the SETUP WIRES SETUP screen and the SETUP FLAGS SETUP screen to change user tool and user frame assignments.

The threshold tests are not active. Use the SETUP THRESHOLD screen to activate and set limits for these tests.

The default robot group number is 1. TO change the group number for multi group controllers, use the SETUP TORCH SETUP screen and change the Group No.

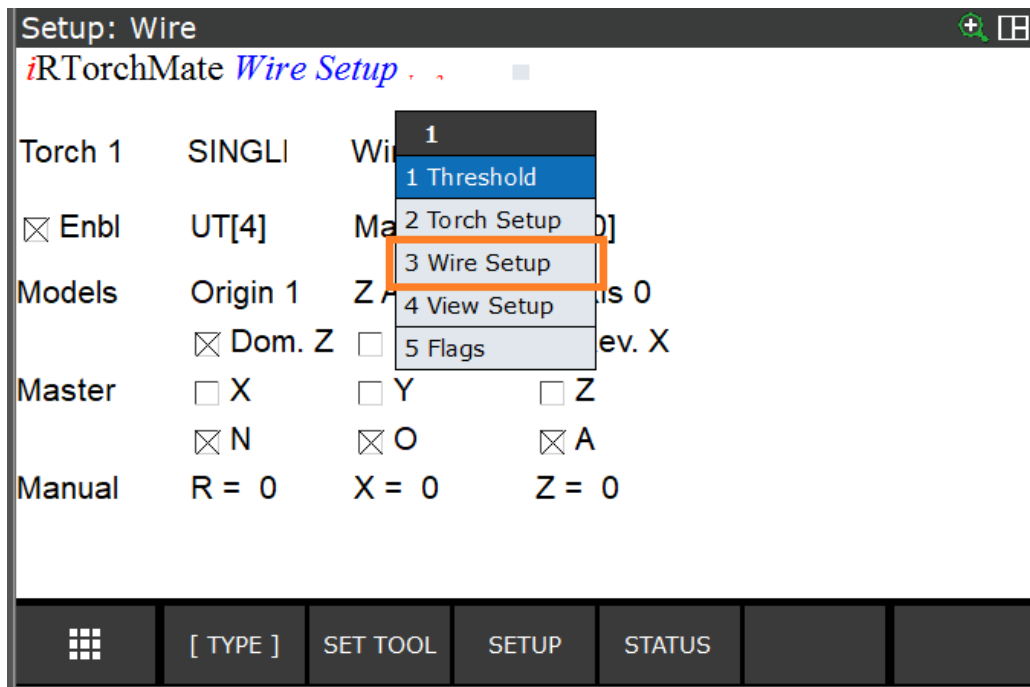


Figure 6-4 Navigating to the Wire Setup screen

By default, the Threshold tests are disabled (not part of this setup).

They can be enabled later once the system is setup. If the first torch is a single wire torch then the Wire Setup (Figure 6-5) is probably correct.

Change the UTOOL numbers if you want to use different tool numbers then change UT[4] for the Single wire tool tip and UT[8] for the master tool tip for the single wire.

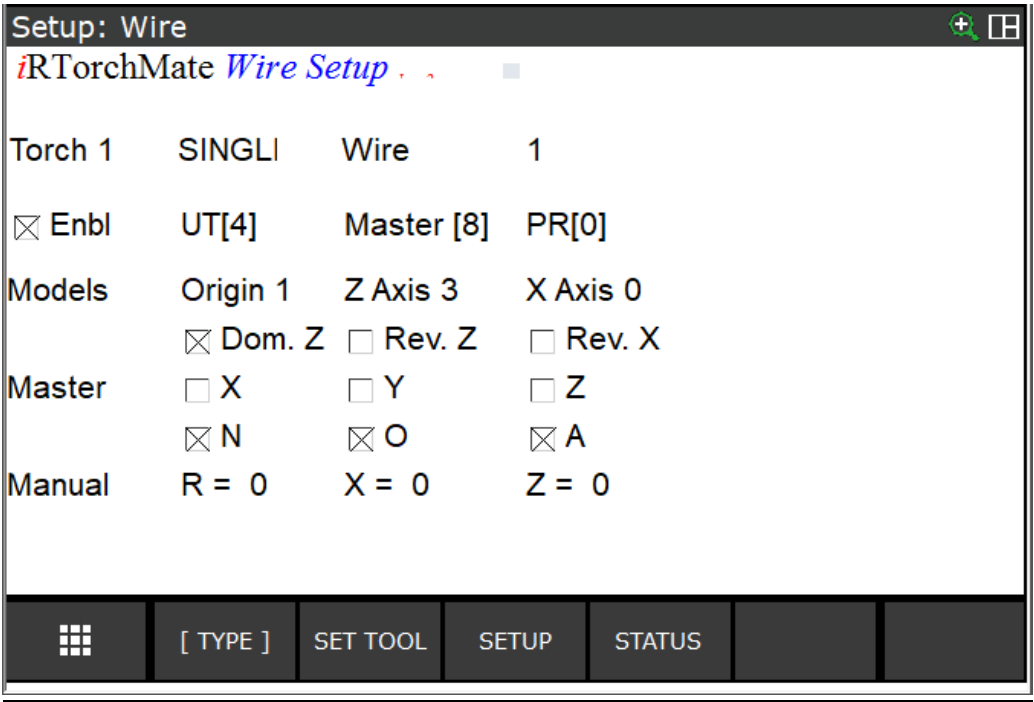


Figure 6-5 Wire Setup Screen

6.4.4. Modifying Lamp Control

Navigate to the Control Flags screen (Figure 6-6) by selecting [SETUP] and then the FLAGS option. The Control Flags screen is displayed. If the user wants to control the lamp output of the lamp in the inspection box, it should be assigned to the Lamp DOUT[x] field.

The Check Box control is disabled until a valid DOUT index is entered.

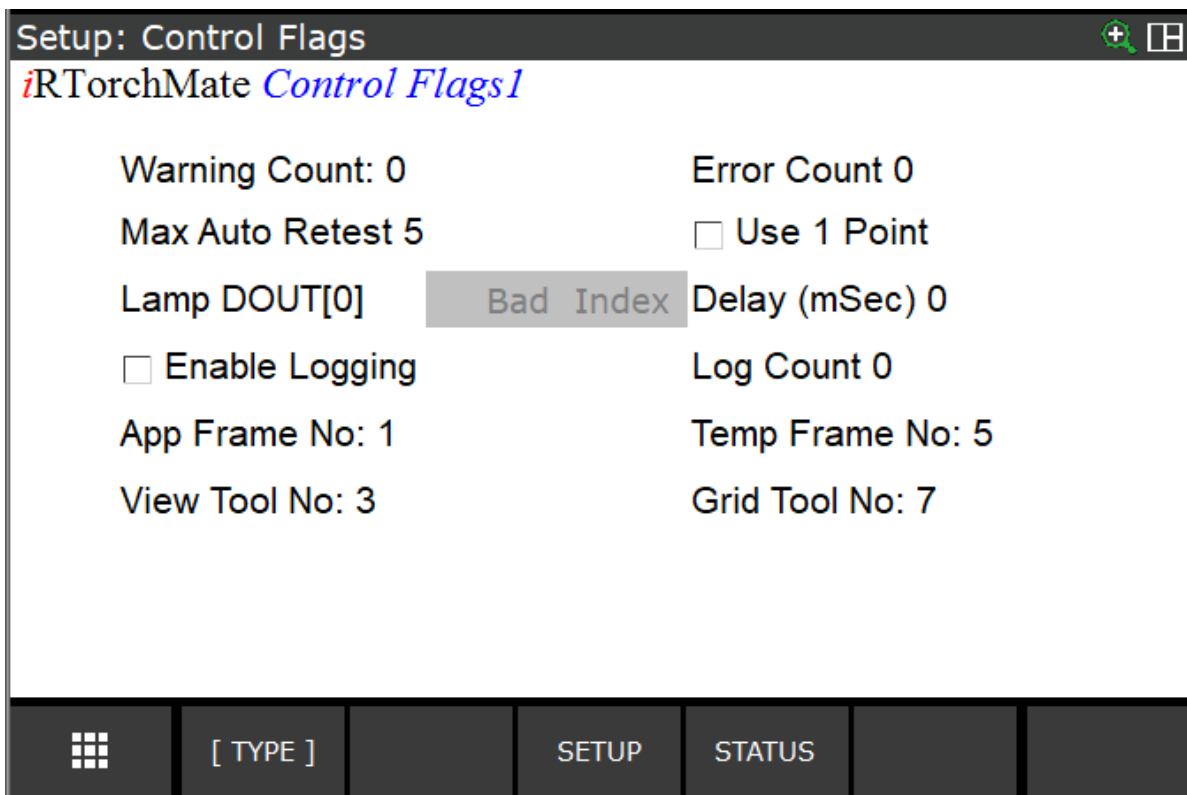


Figure 6-6 Control Flags Screen

6.5. Setup Vision Data

The vision data contains factory default values. The camera data is probably correct. The camera calibration setup will need to be recalibrated. The Vision process data may require adjustment to the search area.

6.5.1. Setup Camera Data

The name of the camera data file is *IRTCALIB_T1*. The user has to setup the Camera setup screen and the Camera calibration screen. Camera data file is loaded when the software is installed. It defaults to port #1. In most cases no modifications of the Camera Data are required. You might need to adjust the focus and aperture of the camera. The default settings for camera data in *IRTCALIB_T1* are shown in Figure 6-7.

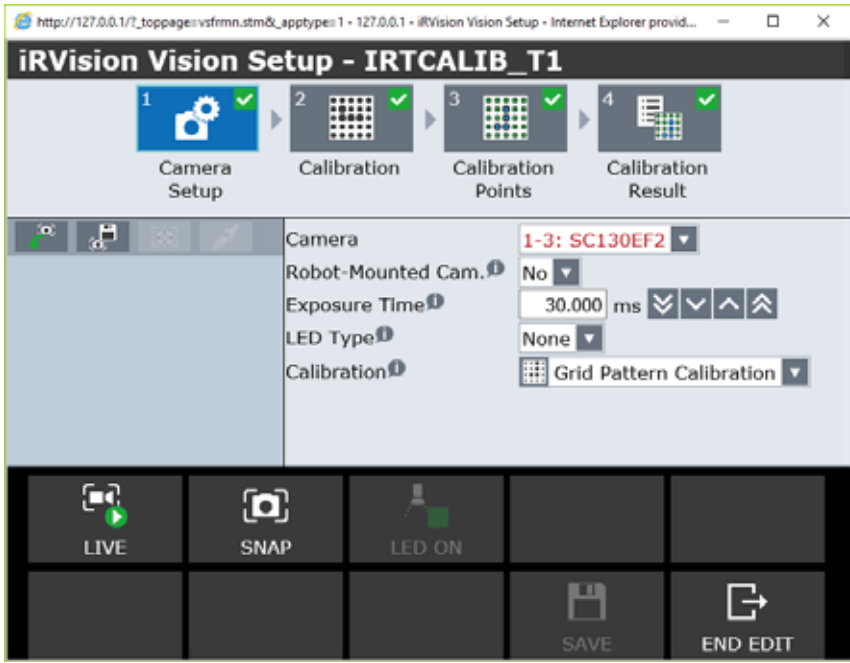


Figure 6-7 iRVision Camera Setup Screen

If the camera listed in the “Camera” field is different from the camera that is connected to the robot, it will be displayed in “red” as is the case in the above figure. Select the correct camera that is connected to the robot from the drop down list beside the field. The following screen will be displayed.



Figure 6-8 iRVision Camera Setup Screen

This is asking the user to use the previously calibrated data. Press Cancel to use the previously calibrated data. Save the settings. The aperture and focus of the camera should be adjusted after the calibration grid is attached to the tool and with the Camera Setup screen in LIVE mode. For best results, jog the robot so the middle of the FOV (large corner circle) is overlaid by the blue cross hairs.

6.5.2. Attach the grid paddle to the torch.

This step is required before the vision camera setup can be completed.

Attach the Calibration Grid Paddle to the Torch using the supplied hose clamp. The paddle is a piece of Plexiglas with an attached 5.0 mm calibration grid. Teach this tool as *GRID* tool. By default this is assigned to Tool 7. The Calibration Grid mounted on the paddle is shown in Figure 6-9.

Attach the paddle to the torch using the clamp. Note: this is a temporary setup.

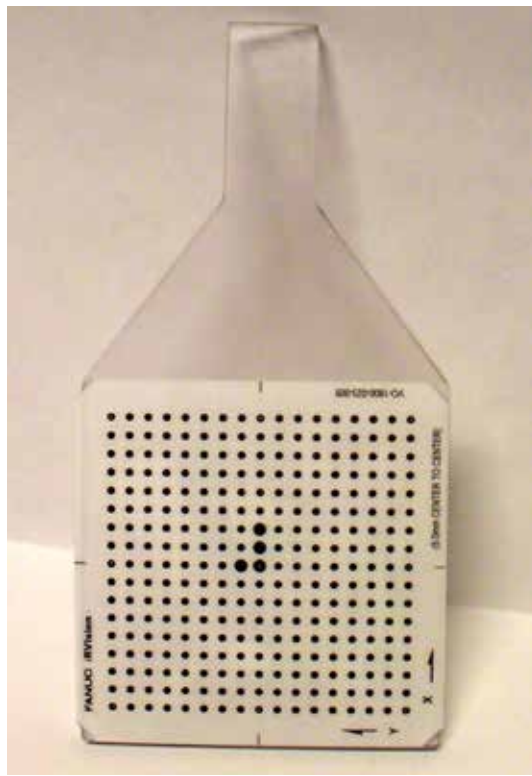


Figure 6-9 Calibration Grid Paddle

Grid Tool, the user tool assigned to the calibration grid can be taught using the conventional six point method or by using the *iRVision* Automatic Grid Frame method. If training manually, make sure the tool origin is the large corner circle. The X and Y direction are marked on the Grid.

Since the accuracy of the calibration grid is most important, it is recommended that the user trains the tool using the second method. It is very difficult for an advance robot programmer to obtain better results. Note that the *iRVision* Automatic Grid Frame method always uses camera in port #1. If the camera assigned to the inspection box is not Port #1, then swap the cables when using Automatic Grid Frame utility.

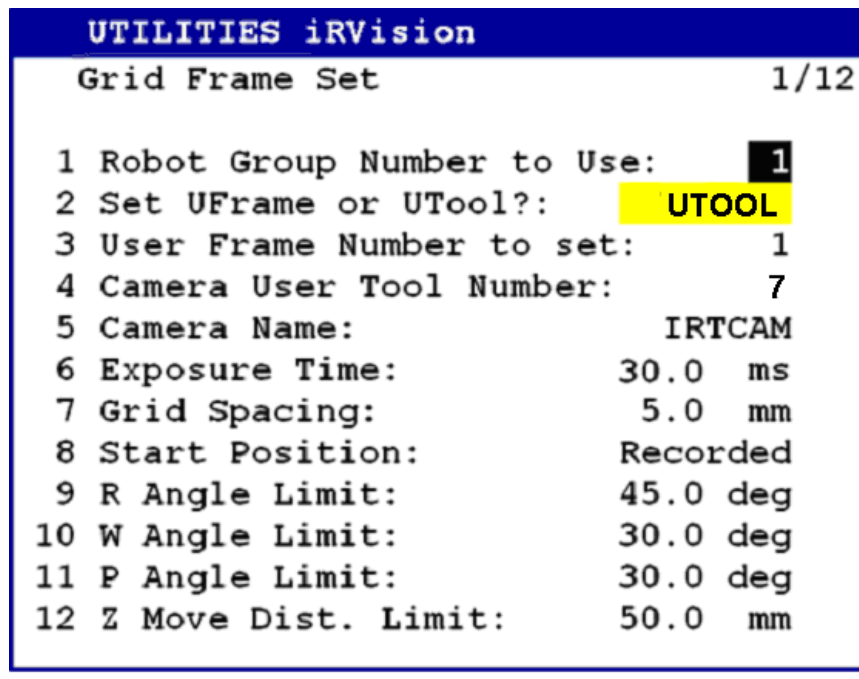
6.5.3. Teach Grid Tool as the Calibration Grid Tool.

This section applies only to the iRVision Grid Frame Set method. Remove the camera from the enclosure and mount it where there is enough room for the robot to manipulate the grid during automatic motion.

Hint: Using a camera tripod simplifies the operation. Choose a camera (tripod) location where the robot has good reach. The motion pattern of the robot resembles a satellite dish with the camera located at the signal focal point and the robot moving the paddle so that the face is normal to the signal focal point. There are approximately 27 positions evenly spaced along the dish.

Use iRVision Automatic Grid Frame Set option (see Chapter 11 of the *iRCalibration Operators Manual*) to teach *GRID TOOL* (default is UTOOL 7). The recommended setup values are shown in Figure 6-10.

Items 1, 2, and 4 (defaults) need to be set to 1, UTOOL, and 7.



The screenshot shows a terminal window titled "UTILITIES iRVision" with a subtitle "Grid Frame Set" and a page indicator "1/12". It lists 12 configuration items, each with a number, a label, and a value. Item 2, "Set UFrame or UTool?:", has the value "UTOOL" highlighted in yellow. Item 1, "Robot Group Number to Use:", has the value "1" highlighted in black.

Item	Label	Value
1	Robot Group Number to Use:	1
2	Set UFrame or UTool?:	UTOOL
3	User Frame Number to set:	1
4	Camera User Tool Number:	7
5	Camera Name:	IRTCAM
6	Exposure Time:	30.0 ms
7	Grid Spacing:	5.0 mm
8	Start Position:	Recorded
9	R Angle Limit:	45.0 deg
10	W Angle Limit:	30.0 deg
11	P Angle Limit:	30.0 deg
12	Z Move Dist. Limit:	50.0 mm

Figure 6-10 iRVision Grid Frame

Set default values. You are required to train only one position for automatic setup. With the camera setup in LIVE mode, record the position so that the image looks like the one in Figure 6-11.

The grid may appear smaller depending on the distance from the camera. 400 mm (16 inches) is a good camera to grid paddle distance. It is OK to change the focus. Remember to restore the focus when the camera is placed back in the Inspection Box.

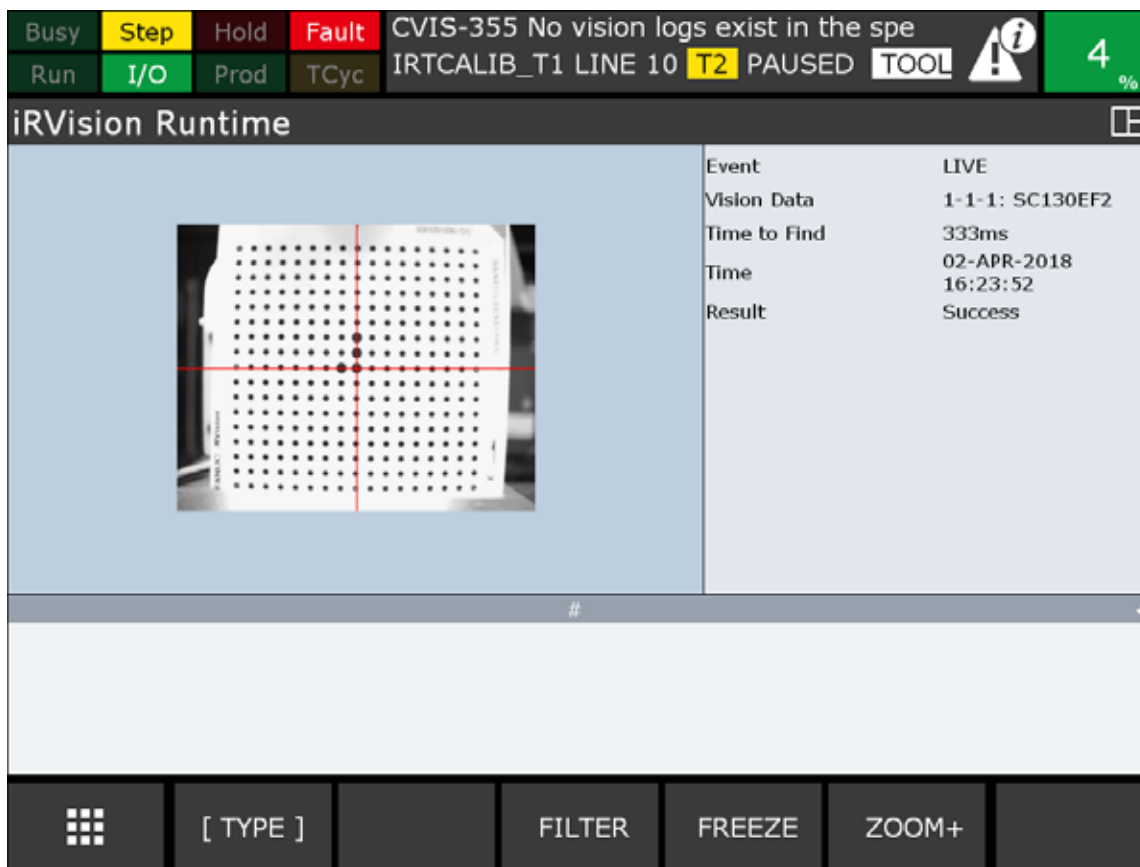


Figure 6-11 Blue Cross Hair in LIVE mode

After the *GRID TOOL* has been taught, confirm the accuracy of the tool by jogging the robot with the tool active and seeing the cross hair in “live” camera mode. After confirming the *GRID TOOL* accuracy, return the camera to *iRVision TorchMate* enclosure. Note the following for setting up the camera focus and aperture

- Loosen the set screws on the focus and aperture rings so the Focus and Aperture can be adjusted.
- Dial the aperture down so the individual LEDs are just barely recognized.
- Slowly open the aperture so the LED outline just disappears.
- Verify the image is in focus.
- Lock the set screws on the focus and aperture ring.

6.5.4. Teach the User Frame of the Inspection box

Enable the Calibration Grid Tool. Jog the robot so the grid is centered in the camera FOV and at the correct distance. Distance from the camera should be in line with the two tapped holes on base in front of white diffuser. Adjust the positions orientation so that the grid's large circles are on the blue cross hair lines of the camera's LIVE image. The dominant axis is the Z axis. It should be align with the Camera line. The LIVE image should also look like Figure 6-11.

IRTSETFRAME_T1.TP is a template program used to train the User frame. The relative locations of P[1] through P[3] are shown in Figure 6-12. These positions represent the origin X axis and Z axis of User Frame 1.

If you choose to RUN IRTSETUP_T1.TP (highly recommended) then let it call irtSetFrame_T1 for you. Otherwise enter the IRTSETFRAME_T1 program and follow the steps listed below.

1. Enable UTOOL 7 and Open IRTSETFRAME_T1.TP.
2. Execute lines 1, 2 and 3 of IRTSETFRAME_T1.TP
3. Touch up P[1], [2], and [3].
4. Teach UFRAME origin P[1] so large corner circle is on cross hair (see Figure Blue Cross Hair in LIVE mode)
5. Teach UFRAME X axis P[2] by jogging robot upward (tool X) ~50 mm. Adjust so that smaller circle in the last row directly beneath large corner circle is on the vertical blue cross hair line.
6. Move back to Origin.
7. Teach UFRAME Z axis P[3] by jogging grid toward camera (tool +Z) ~50 mm and adjust so large corner circle is still on cross hair.
8. Run IRTSETFRAME_T1.TP

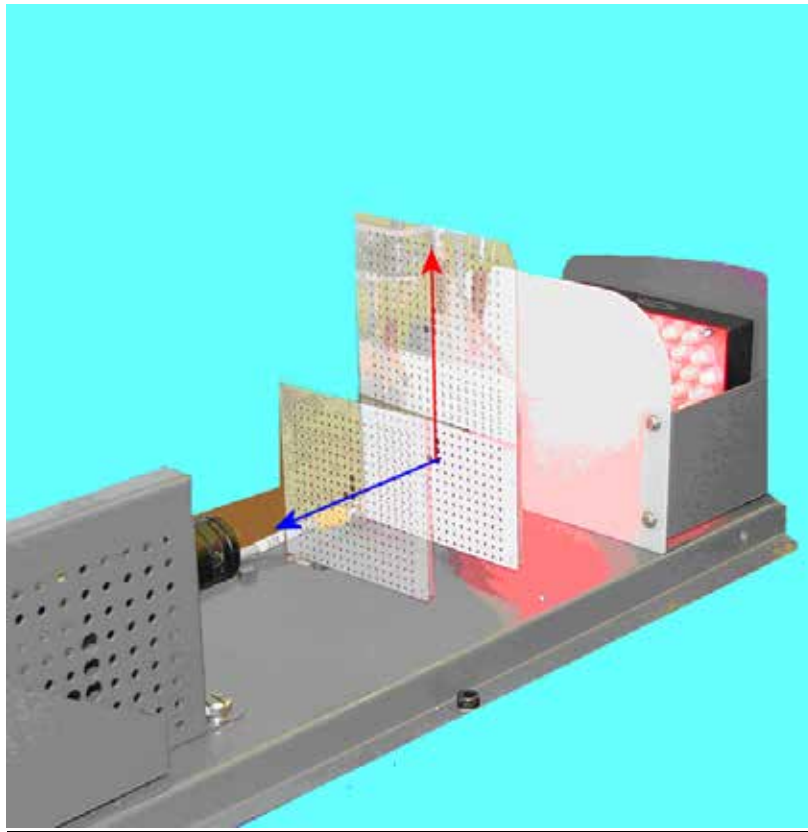


Figure 6-12 Relative position of SETFRAME.TP P[1], P[2], and P[3].

After the program is run, a user frame is calculated using functions in *iRMath*. The resulting frame (User Frame 1) will have the coordinate system displayed in Figure 6-13 . (Note Inspection box is shown rotated 180 degrees from Figure 6-12).

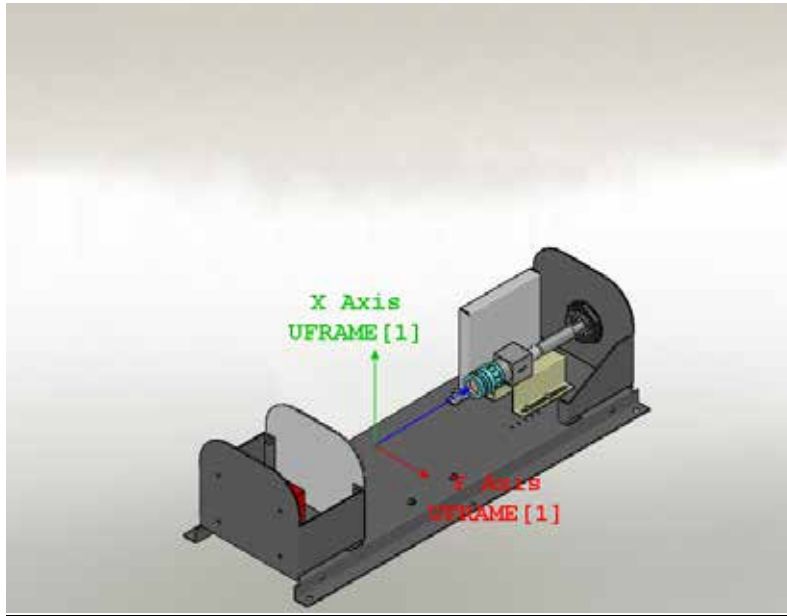


Figure 6-13 Application User Frame of Inspection Box

Verify the User application frame (inspection box frame) is correct. With the robot at the origin position, UFRAME NUM set to *APP FRAME* (default = 1), and UTOOL NUM set to *GRID TOOL* (default = 7) look at the Teach Pendant's Position screen in User frame mode. The position should be very near to (0,0,0,0,0,0). This is a very convenient setup for verifying correct tools and frames. Whenever a UTOOL is active in User frame 1 and at Position (0,0,0), the tool tip should be under the LIVE cross hair.

6.5.5. Perform Camera Calibration

Set up the calibration using Application User Frame 1. The camera is fixed and the calibration fixture is attached to the robot (UTOOL 7). The user should use the IRTCALIB_T1.TP program for calibrating. Note that with the Grid Tool attached and UTOOL set to *GRID TOOL* (default = 7) and UFRAME set to *APP FRAME* (default 1), the Center of the Cross hair should be (0,0,0). The default positions for P[1], P[2], and P[3] may work without touchup. However, there is the possibility that the taught positions were taught using a different type of robot than what the user is currently using. In this case, the Configuration string of the positions will not match and should be recorded again.

Hint: To easily retrain the positions for a new configuration, record the current position and manually reset P[1] to [0,0,0,0,0,0], P[2] to [0,0,25,0,0,0], and P[3] to [0,0,-25,0,0,0].

P[1] is the origin of the user fame. P[2] is the view position of first calibration Plane, +25 mm in Z from origin. P[3] is the view position for the second calibration plane, -25 mm from the origin. Use the values for Camera Calibration Setup as shown in Figure 6-14.

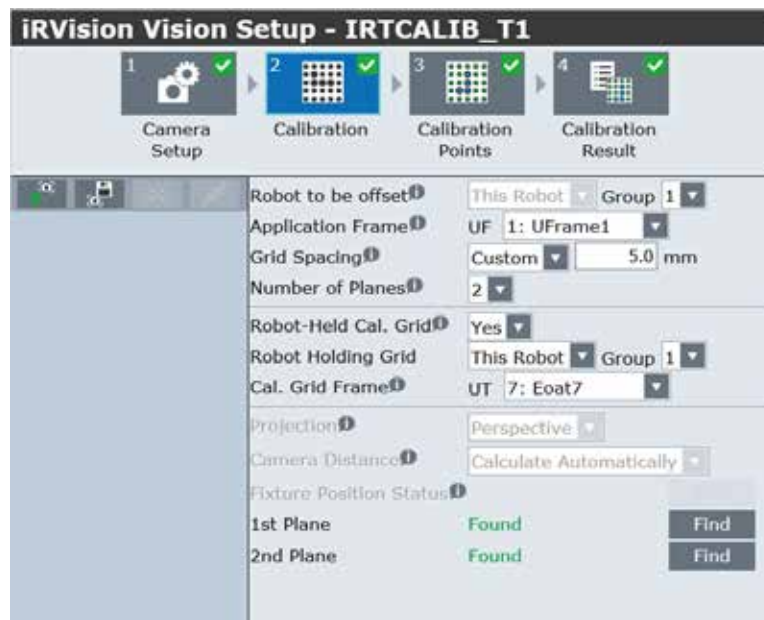


Figure 6-14 Default values for Grid Calibration (IRTCALIB)

Follow these steps

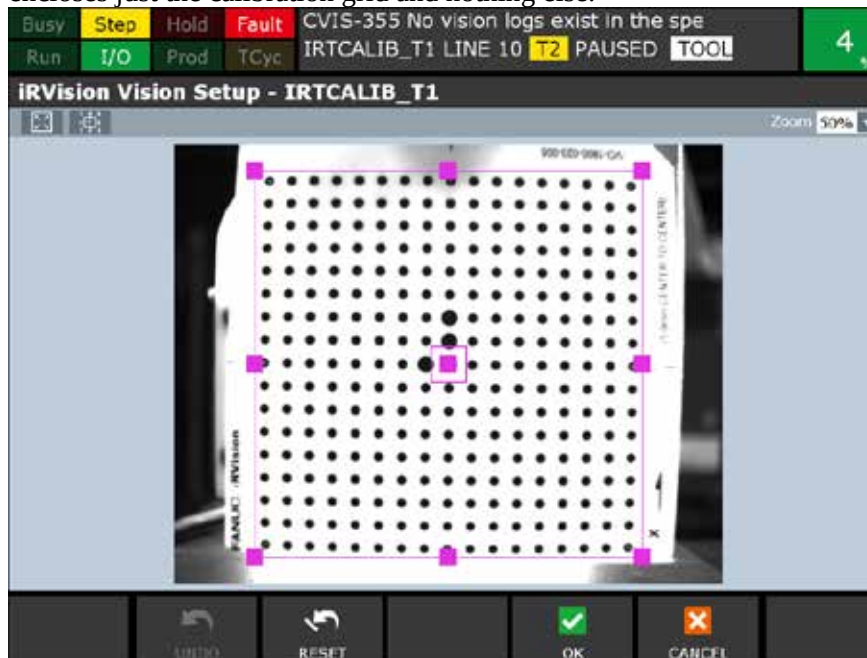
1. Open the camera data tool setup page.
2. Go to camera calibration setup screen
3. Select IRTCALIB_T1, ENTER
4. Edit to default values as shown above, save and close. Note: The user must exit this screen before running IRTCALIB_T1. Otherwise an error is posted and it does not calibrate. The remedy is to close the screen and run the program again.
5. Run IRTCALIB_T1.TP (This program will calibrate the camera, run at 5% speed.)
6. Remove Grid from Torch

In the event of any errors during this calibration, the user must recalibrate the camera. Follow these steps:

1. Jog the robot to P[2]. Click Snap.
2. In the Calibration tab of *IRTCALIB_T1*, press “Find” against the “1st Plane”.



3. The following screen will open up. Adjust the pointers such that the rectangular selection box encloses just the calibration grid and nothing else.

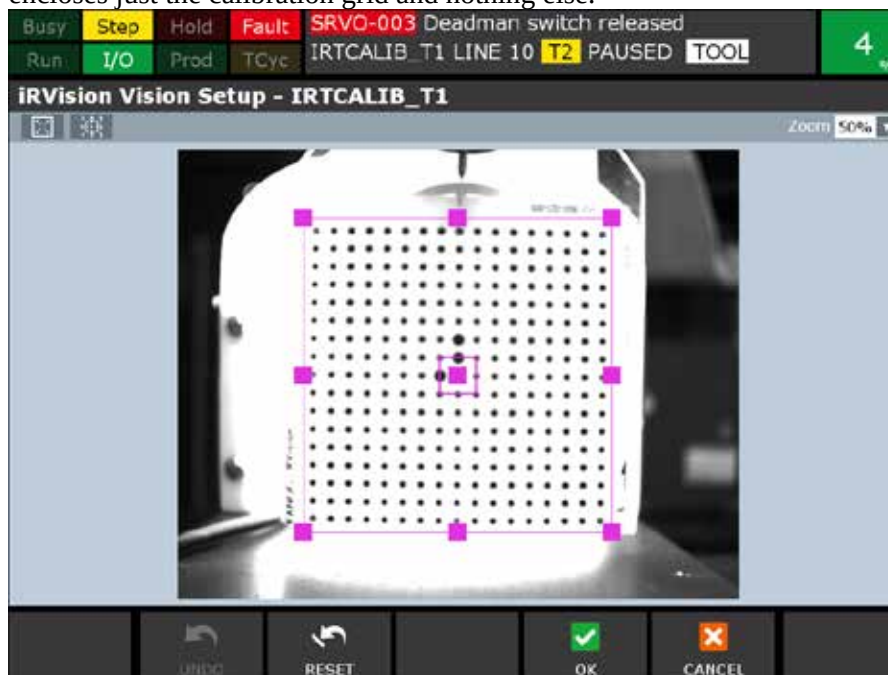


Note: The individual squares of the rectangular box can be selected by the numbers on the iPendant and then moved with the arrow keys.

4. Press ok. If the calibration is not successful, repeat the steps 2 and 3.
5. Jog the robot to P[3]. Click Snap.
6. In the calibration tab of IRTCALIB_T1, press “Find” against the “2nd Plane”.



7. The following screen will open up. Adjust the pointers such that the rectangular selection box encloses just the calibration grid and nothing else.



8. Press ok. If the calibration is not successful, repeat the steps 6 and 7.

6.5.6. Teach Master UTOOL(S)

If the IRTSETUP_T1 is run, this section can be skipped.

Using 6 point method, Teach the original / Master TCP values to the respective positions as shown below.

Figure 6-15 shows the comment field for each tool or how the tools are assigned. The X, Y, and Z values are not the expected tool values.

SETUP Frames					
Tool	Frame	/ Direct Entry			1/10
	X	Y	Z	Comment	
1	0.0	0.0	0.0	[UPDATED DUAL TOOL 1]	
2	0.0	0.0	0.0	[UPDATED DUAL TOOL 2]	
3	0.0	0.0	0.0	[UPDATED VIEW TOOL 3]	
4	0.0	0.0	0.0	[UPDATED SNGL TOOL 1]	
5	0.0	0.0	0.0	[EOAT 5	]
6	0.0	0.0	0.0	[EOAT 6	]
7	0.0	0.0	0.0	[GRID UTOOL 1	]
8	0.0	0.0	0.0	[MASTER SNGL TOOL 1	]
9	0.0	0.0	0.0	[MASTER DUAL TOOL 1	]
10	0.0	0.0	0.0	[MASTER DUAL TOOL 2	]

Figure 6-15 UTOOLS

6.5.7. Set up Vision Process

Actual time to perform the inspection depends on how fast the robot can travel to the inspection station, enter and exit the station, and which inspections are performed. At a 50% override speed, it is less than 5 seconds.

There is one vision process provided as part of the software. This Vision process is used for both torch types. These can be viewed in the View Setup Screen.

Vision Process Name	Description
IRTWIRE_T1	View Position 1 and 2, Single & Dual Wire Torch's

Table A-1 Vision Processes Table

The next chapter discusses the vision process.

7

VISION

7. VISION

7.1. Overview

Three pieces of vision data are required to run a vision program. The Camera data describes the camera setup. The Process data defines the vision program. This section describes each type of data.

7.2. Camera Data

Included and installed with the software is a camera data setup file called IRTCALIB_T1.VD. This camera data setup is installed with factory default values that do not need to be modified. The camera data setup file has two main setup pages. One is Camera Setup Page and the other is Camera Calibration Page.

The Camera Setup page identifies the type of camera and the port number. It also defines if the camera is mounted on a robot. iRVision TorchMate uses the KOWA SC130EF2 camera. The Field of View contains 480 by 512 pixels. The camera is not mounted on the robot. It is in a fixed position inside the Inspection Box. There is one camera per inspection box.

The Camera Calibration setup page's main purpose is to translate the vision image row and column location into a position in the application user frame. The camera calibration has a default calibration setup. This is close to expected calibration but not accurate enough to use. The camera calibration data will need to be recalibrated once the camera has been fixed in the Inspection Box and the Inspection Box has been placed in its final position.

There is one camera data tool for each robot group that will use the Inspection Box. To calibrate a camera, the location of the application frame and the location of the calibration grid must be defined. Our calibration setup uses the user frame of the Inspection box as the Application User Frame. The grid is affixed to the torch using the paddle grid and taught as a user tool.

Training the application frame, training the calibration grid tool, and calibrating the camera have been automated. It is controlled by TP programs. IRTCALIB_T1.VD contains the camera calibration data as well.

7.3. Vision Process Setup

The vision process defines which vision tools are used to find the target. With iRVision TorchMate many targets are found along the wire. All the points are passed to a KAREL program which uses the data to calculate the UTOOL of the wires.

A vision process named IRTWIRE_T1 is included with the installations. Generally only the location of the search window needs to be modified. This vision process has one *Edge Point* tools. The Model ID should be 1. It contains a search window that may need to be re-positioned..

7.3.1. Train the Vision Process

The purpose of this vision process is to find the angle of the wire and the tip of the wire. A dual wire torch is shown in multiple points along the wires are found displayed as green dashes.

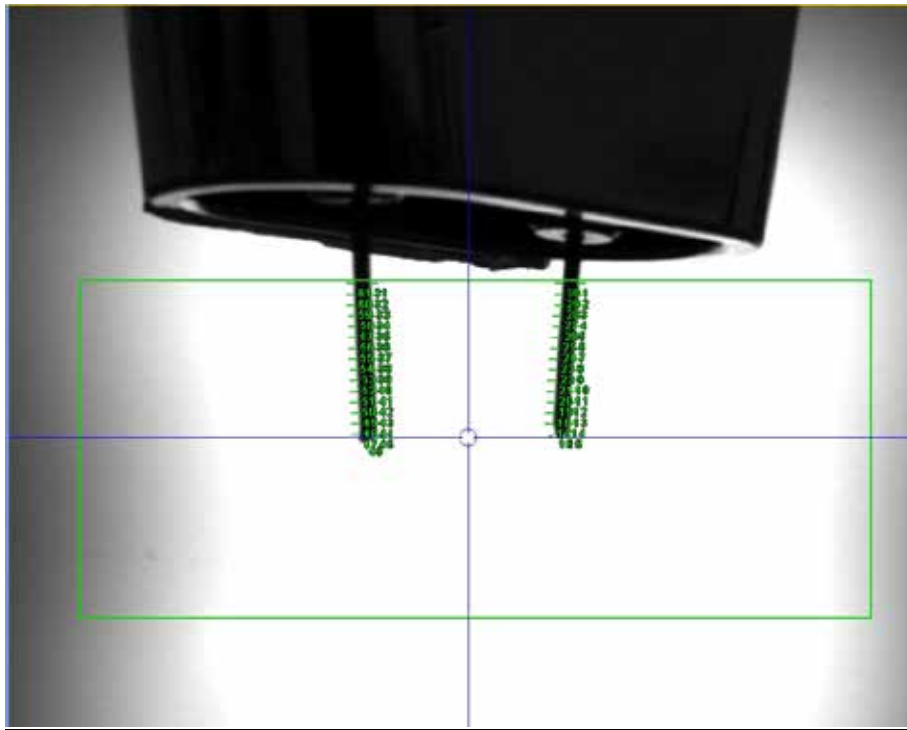


Figure 7-1 Dual Wire Training Image

To find these targets an Image To Points Vision Process is created with a Selected Edge Points Locator tool. Figure 7-2 shows the vision tree for the vision process.

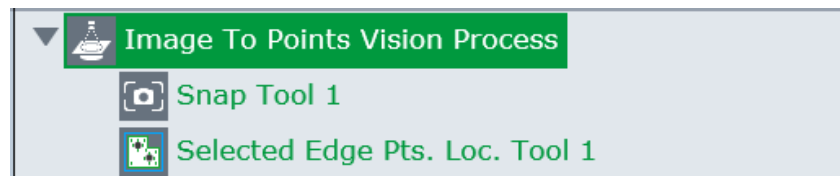


Figure 7-2 IRTWIRE Vision Process

The second image (Figure 7-3 shows an alternative method. In this case the user wants to shift the search area of the Selected Edge Points Location tool by using the GPM Locator Tool to first find the nozzle of the torch and then shift the second tools search area. This would only be required when the tool nozzle moves between tests.

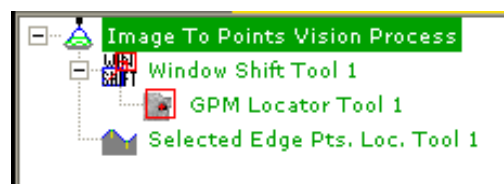


Figure 7-3 IRTWIRE Vision Process

Train the vision process to have the default settings as shown in Figure 7-4.

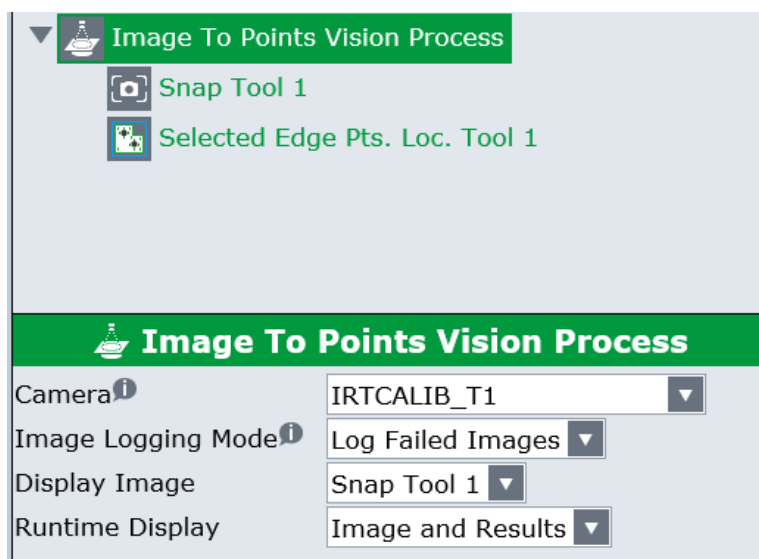


Figure 7-4 2-D Vision Process

7.3.2. Setup the Snap Tool

Setup the snap tool to have the values as shown in Figure 7-5.

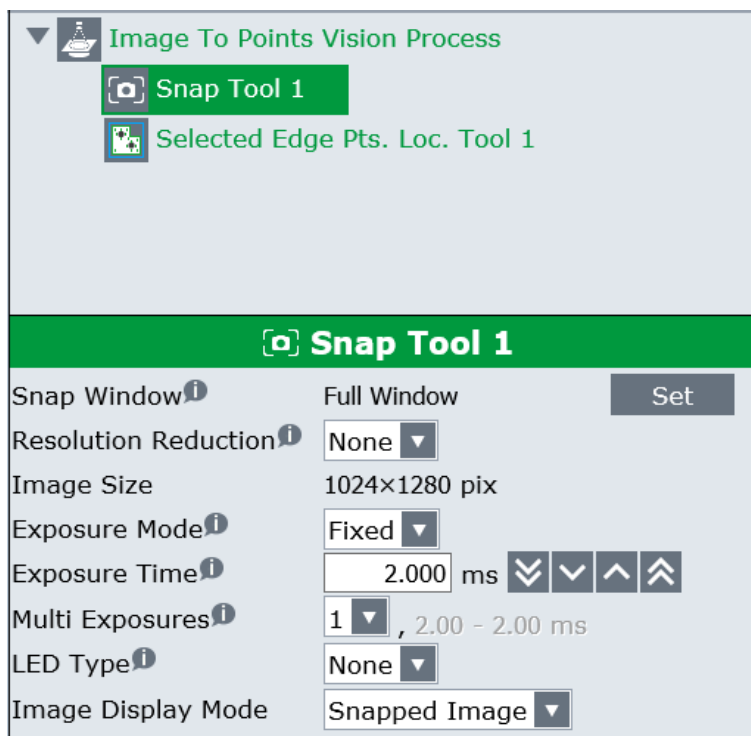


Figure 7-5 Snap Tool

7.3.3. Train the Selected Edge Points Tool

The Selected Edge Points Tool is used to find the bottom of the wire. Select the Edge Trace Tool from the list by double clicking the item. In Figure 7-6 the Selected Edge Locator Points Tool 1 is highlighted when selected.

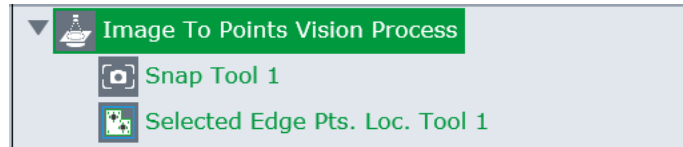


Figure 7-6 IRTWIRE T1 Image To Points Vision Process

7.3.4. Set up the Selected Edge Points Tool

Set the default items as defined in Figure 7-7. Make sure the Starting Model ID is 1.

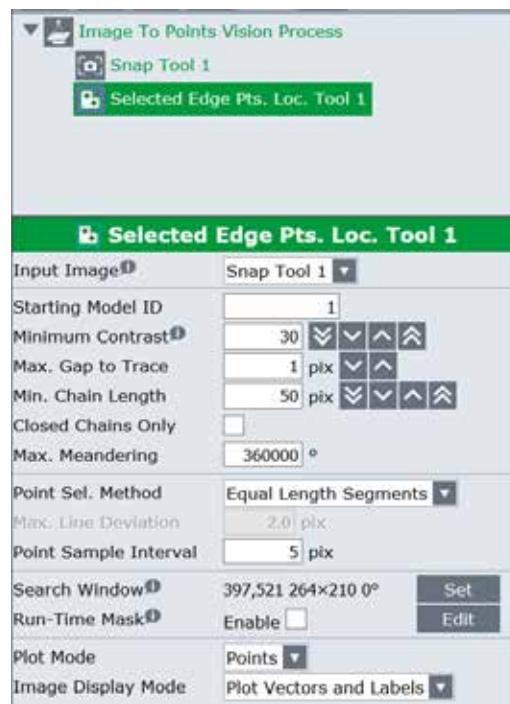


Figure 7-7 Selected Edge Points Tool Setup

7.3.5. Set the Search Window.

The search window is defined by clicking the Set Search Window button. Once selected a red box is moved until it encloses the desired search area.

**WARNING**

Do not include any of the tool head in the search window.

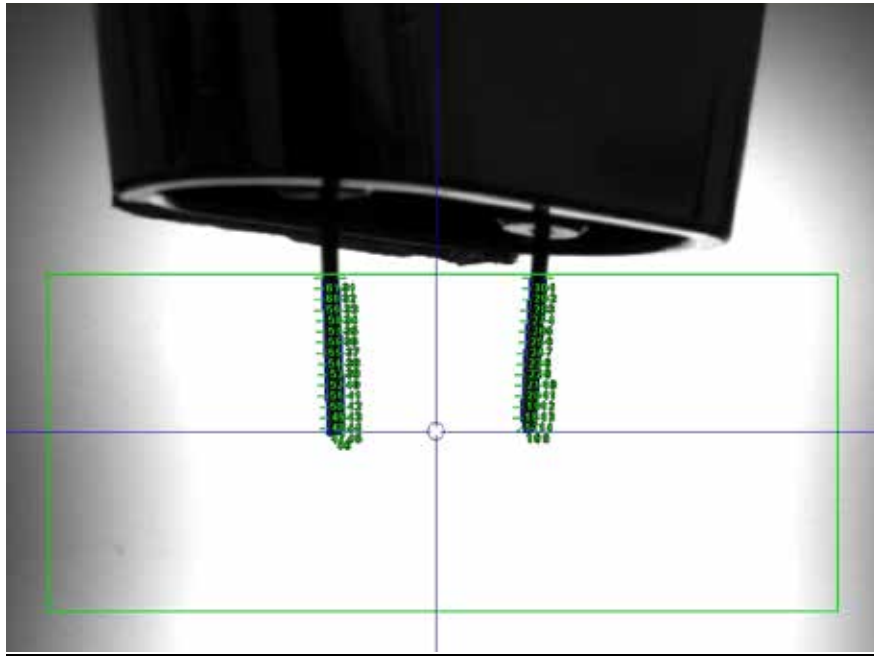


Figure 7-8 Selected Edge Points Search Window for Dual wire

7.3.6. Test the Vision Process

Test the process by pressing the **Snap + Find** button. The image should look similar to the single wire in Figure 7-9. The Dual wire is shown in Figure 7-8.

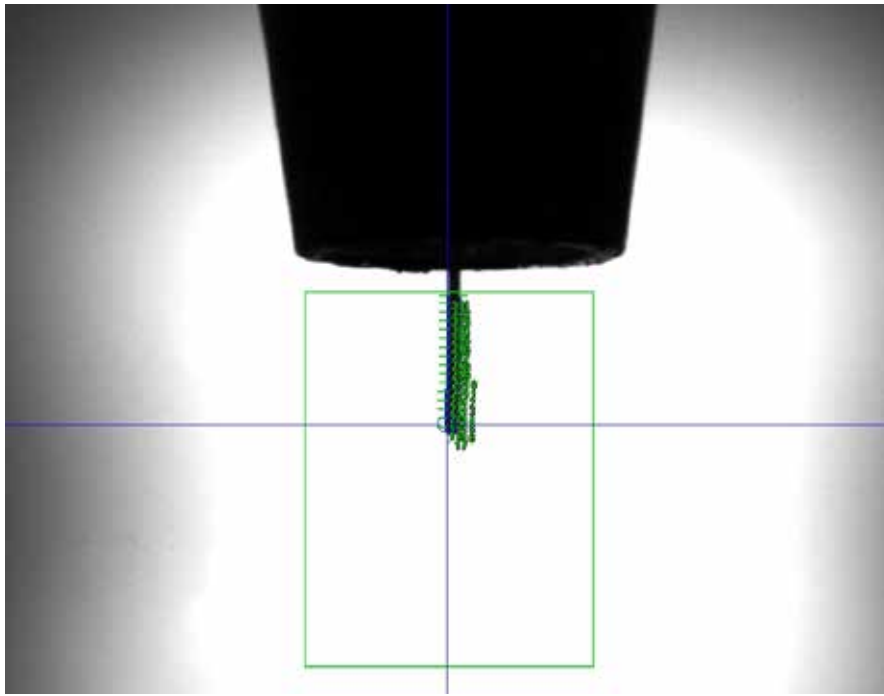


Figure 7-9 Image after Snap and Find of a single wire

Multiple positions will be found as shown in the Found Results Table in Figure 7-10. The slide bar can be used to show all the points.

Camera										Snap and Feed		Continuous S++		Feed	
Found Results															
		X	Y	Bull	Model ID	Score	Comment	Fx En							
Found:	32	1	-0.2	-0.5	-65.2	1	100.0	214.0	0.000						
		2	-0.1	-0.5	-64.7	1	100.0	195.0	0.000						
Time to Find:	91 ms.	3	0.3	-0.2	-67.4	1	100.0	193.0	0.000						
		4	0.4	-0.8	-81.2	1	100.0	197.0	0.000						
Time Found:		5	0.5	-0.7	-88.0	1	100.0	192.0	0.000						
		6	1.0	-0.8	-89.8	1	100.0	217.0	0.000						
		7	1.5	-0.2	-89.9	1	100.0	199.0	0.000						
		8	1.6	-0.8	-82.4	1	100.0	196.0	0.000						
		9	2.1	-0.2	-88.0	1	100.0	192.0	0.000						

Figure 7-10 Found Results

8

GRAPHIC USER INTERFACE

8. GRAPHIC USER INTERFACE

8.1. Overview

The robot user interface consists of two sets of screens, the setup screens and the status screens. The IRT Status screen can be activated when viewing any screen by simply pressing the Status key on the Teach Pendant. Press the [TYPE] F1 key and choose IRT Status from the pull down menu.

In general you will probably only use the STATUS and the Tool Setup screens. The other screens are used for customizing the application or trouble shooting

The user can navigate between iRVision TorchMate screens using the TP Function keys. Press the F3 function key for the Setup menus, or the F4 function key for the Status menus.

8.1.1. Setup Menu

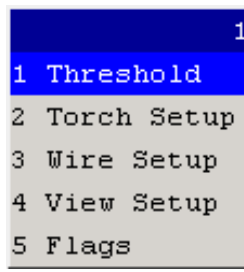


Figure 8-1 Popup Setup Menu

The popup Setup menu includes

Threshold Use this screen to enable weld tip warning and error thresholds and assign Digital Outputs for error handling.

Torch Setup Use this screen to set number of wires, Group number and LL threshold.

Wire Setup Use this screen to set robot dominant axis per tool and set update masking per tool.

View Setup Use this screen to set vision process and storage location of focal point.

Flags Use this screen for data logging and setting nominal TCP values.

8.1.2. Status Menu

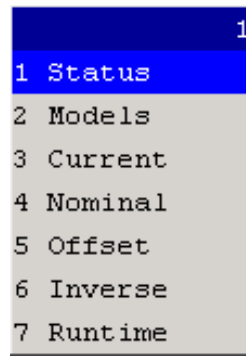


Figure 8-2 Popup Status Menu

The popup **Status** menu includes

Status Use this screen to display current test status, and measured data for each test.

Models Use this screen to display the vision models

Current Use this screen to display the current TCP values

Nominal Use this screen to display the TCP nominal values

Inverse Use this screen to display the inverse TCP values

Runtime Use this screen to view the vision process display

8.2. IRVision TorchMate Threshold Screen

Enable or disable a threshold test by clicking on the Post Warnings or Post Errors label. The test is enabled when box has an 'x' in it.

After a UTOOL has been calculated it is compared to the XYZWPR of the Nominal UTOOL. The difference is compared to the threshold values. If the absolute value exceeds an error value, an error is posted. If the difference exceeds a warning threshold, a warning is posted.

If the threshold value is set to 0, no test of that element is executed. This is equivalent to setting the threshold to a very high value.

If an error is posted, the system pauses for operator recovery.

Setup: Threshold	
<i>iRTorchMate Threshold Setup</i>	
☐ Warnings	☐ Errors
DO[0]	DO[0]
X: 5	X: 0.001
Y: 5	Y: 10
Z: 5	Z: 10
W: 5	W: 10
P: 7	P: 10
R: 5	R: 10
[TYPE]	SETUP STATUS

Figure 8-3 Threshold Setup

8.3. IRVision TorchMate Torch Setup Screen

This screen is used to define the robot group number, the number of wires in the torch, and the Line to Line Threshold distance.

Setup: Torch	
<i>iRTorchMate Torch Setup</i>	
Torch Index 1	Group No 1
Number of Wires 1	P2W Threshold 5
Local View Tool [0]	
☐ Multi Pass	☐ Calc Fit Error
[TYPE]	SETUP STATUS

Figure 8-4 Torch Setup

The software supports four torches. With a multi arm controller, a separate torch can be assign to each group. When the user has less than four robot groups, the user can assign the torch group number to any group.

Torch Index is the number of the torch. Enter 1 to 4. Normally the torch number 1 is assigned to robot group 1, etc...

Group No is used to assign a robot group number to a torch.

Number of Wires is the number of wires on a torch. Enter 1 or 2.

P2W Threshold is the maximum allowable distance a point can vary from the center line. Points greater than this (pixel) distance are discarded. After the points are discarded the center line is recalculated. The default value is 5 pixels. This is a good value for wire. If the target is not a wire, the value should be increased to include the desired points. The purpose of this threshold is to toss points that may be part of the torch head and not the wire. This may occur if the view position is taught too low or the Search Window is taught too high.

Local View Tool [x] is an alternative view number. If your work cell supports a gun changer, then it is possible that a second torch will have a different orientation when mounted on the face plate then the original torch. When the value is 0, the View Tool assigned in the SETUP FLAGS screen will be used. If this value is non zero, then this tool number is used instead of the SETUP FLAGS screen view tool

Multi Pass flag indicates if some data points on the wire should be removed from the calculation for best wire fit. Points too far (**as defined by P2W Threshold**) from the best line fit wire are remove from the set of data and the best line fit is performed a second time. The original intention was to throw out points that are clearly not on the wire. This operation does take time so it should be excluded when not needed.

Calc Fit Error flag determines if the fit error is to be calculated. This operation does take time so it should be excluded when not needed.

8.4. IRVision TorchMate Wire Setup Screen

This screen is used to define how the targets are used to define the UTOOL of the wire. To define a wire the user selects three models, the origin, the X axis, and the Z axis. A two wire torch has four models and a one wire torch has two models. Model #1 is point at the bottom of the right wire (for dual torch) or the bottom point for a single wire torch wire (as view in the camera FOV. Model # 3 is the top point on the right wire and inside the search window.

Model #4 is point at the bottom of the left wire (for dual torch). Model # 6 is the top point on the left wire. The X axis and Z axis are defined by the origin model number and the X or Z axis model number. The tool always has a dominant axis. It is the Z axis or the X axis. The reason for a dominant axis is because the user cannot guarantee that the targets used to define the axis are exactly perpendicular. The dominant axis keeps its position and the other axis shifts slight until it is exactly 90 degrees. The targets used are specified by their model number.

Setup: Wire			
<i>iRTorchMate Wire Setup</i>			
Torch 1	Dual	Wire	1
☒ Enbl	UT[1]	Master [9]	PR[0]
Models	Origin 1	Z Axis 3	X Axis 4
	☐ Dom. Z	☐ Rev. Z	☒ Rev. X
Master	☐ X	☐ Y	☐ Z
	☐ N	☐ O	☒ A
Manual	R = 0	X = 0	Z = 0
[TYPE] SET TOOL SETUP STATUS			

Figure 8-5 Wire Setup Screen

The fields are defined below.

Torch (Index) specifies the torch number in use (1 - 4)

Dual or **Single** defines the number of wires in the system. This value can be changed by using the TORCH SETUP screen or the **SET TOOL** button.

Wire (Index) specifies the wire number that is being defined. This value is 1 for a SINGLE wire torch and (1 or 2) for a DUAL wire torch.

Enable (checkbox) this has X in checkbox when enabled.(blank when disabled). If enabled the UTOOL in **UT[x]** is updated at the end of the measurements.

UT[x] specifies the UTOOL that is updated. The tool is modified by changing the x value. In the example screen it is set to 1.

MASTER[x] is used to define default element values of the tool. For example, the user can define which elements are modified and which keep the master values. In the example screen the w & p are set to UTOOL 9.

PR[x] indicates which position register is updated with the tool offset. If **Update MNU** is disabled, then the user may choose to move the robot using the Tool Offset, PR clause. If PR [0] is indicated then no PR is updated.

Dom. Z can be enabled (x in checkbox) when the Z axis is dominant or disabled (blank checkbox) when the X axis is dominant. Note the **Dominant X** field is always the opposite of the **Dominant Z** field.

Rev. Z indicates the direction of the Z axis. When enabled (x in checkbox), the **Origin (Model) to Z Axis (model)** model defines the negative Z axis. When disabled (blank in checkbox), the **Origin (Model) to Z Axis (model)** defines the positive Z axis.

Rev. X indicates the direction of the X axis. When enabled (x in checkbox), the **Origin (Model) to X Axis (model)** defines the negative X axis. When disabled (blank in checkbox), the **Origin (Model) to X Axis (model)** defines the positive X axis.

(Use) Master this allows the user to override the calculated tool with the data from the **master** tool. When **X**, **Y**, or **Z** are enabled then the X, Y, or Z value are over written with the **master** tool values. The **N**, **O**, and **A** refer to the normal, orient, and approach axes (i.e. X, Y, and Z axis respectively).

When an axis is enabled, the user is requesting that the new tool axis is aligned with the **Master** tool axis. For example if the Approach axis is selected then the new tool's XY plane will match the **master** tool XY plane but the rotation may vary. Selecting two axes is mathematically equivalent to selecting all three.

The **Manual** values or **R**, **X**, and **Z** allow the user to add the value to the calculated UTOOL value. The X and Z direction are relative to the new UTOOL X and Y direction. Include a Z value to make the wire longer or -Z to shorten the wire. The Tool X direction can be changed by adding or subtracting R. For example, the XTOOL direction for a single wire is by default is set to align with the face plate (UTOOL[0]). The user can rotate the direction by editing the R value.

SET TOOL (button) is used to reset the default values of the wire. It is located in the F2 function key position. Pressing it displays the following

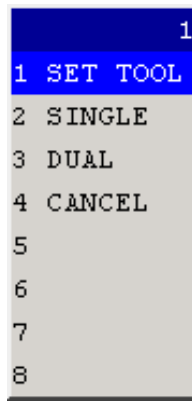


Figure 8-6 Set Tool Button

Single sets all the default values for a single wire torch.

Dual sets all the default values for a dual wire torch.

Cancel returns control to main screen

SET TOOL is a default name that is displayed on the TP at the F2 position when the pop up screen is not active. It functions the same way as Cancel

8.5. IRVision TorchMate View Setup Screen

This screen is used to define names of the vision processes and associate them with the view. The vision processes are called from two robot positions. In general one view is taken at a 45 degree angle to the Inspection Box X axis (up). The other view is at -45 degrees. The two views are 90 degrees apart. In some cases the same Vision process could be used with both views.

Setup: View

iRTorchMate View Setup

Torch Index: 1 View Index: 1

VP Name : IRTWIRE_G1

Frame FP PR [0] Tool FP PR[0]

[TYPE] SETUP STATUS

Figure 8-7 View Setup

The fields are defined below.

Torch Index specifies the gun number in use (1 - 4)

View Index specifies the view number in use (1 or 2)

VR[n] defines the Vision Register to use with the Vision Process.

VP Name refers to the name of the Edge Points Vision Process

Frame FP PR[0] defines the PR that should receive the Focal point of the camera in the Frame coordinates of the User Application Frame when it is non zero. If zero the value is not copied. This is intended for advancing programming use.

Tool FP PR[0] defines the PR that should receive the Focal point of the camera in the Face Plate coordinates of the Robot when it is non zero. If zero the value is not copied. This is intended for advancing programming use.

8.6. (Control) Flags Setup

This screen allows the user to set App Frame number, Grid Tool number, Lighting and delay along with controlling the logging and error/warning counts.

Setup: Control Flags  

iRTorchMate Control Flags1 

Warning Count: 0	Error Count 0
Max Auto Retest 5	☒ Use 1 Point
Lamp DOUT[1] 	Delay (mSec) 0
☐ Enable Logging	Log Count 0
App Frame No: 1	Temp Frame No: 5
View Tool No: 3	Grid Tool No: 7

[TYPE]		SETUP	STATUS		

Figure 8-8 Control Flags

Warning Count displays the number of warnings logged. It can be reset to zero.

Error Count displays the number of errors logged. It can be reset to zero.

Max Auto Retest count defines how many consecutive warnings can occur without generating an error. For example if vision fails to find one of the models, the system will automatically retry. It will continue to retry until this value is exceeded. At this time an excessive error count is generated that requires operator intervention.

Use 1 Point defines how the torch UTOOL is defined in IRTSETUP_T1.TP. TRUE indicates the one point method is used. False indicates the Six Point Method is used. The One Point Method defines the UTOOL as its position relative to the Application Frame when the torch has been jogged so that the torch wire is on the Vision cross hair. Since $F = R : T$ (Frame = FacePlate : UTOOL), the UTOOL can be easily defined as $T = INVERSE(R) : F$.

Lamp DOUT[0] provides an alternative method of controlling the lamp in the Inspection Box. Assigning the DOUT will automatically turn on and off the light when snapped. If the DOUT is invalid or zero, "Invalid" will appear after the field name. If the DOUT value is ON, then a Green circle will appear after the field name. If the DOUT value is OFF, then a Grey circle will appear after the field name. When the DOUT is zero the DOUT is disabled.

Check Box (Invalid) The check box will toggle the DOUT provided it is valid. A port test is conducted on the DOUT index to determine if it is valid. When invalid, the check box control is deactivated and the background is gray. When the check box is valid, the "Invalid" text does not appear and the check box background is white.

Delay (mSec) field defines how long to wait for the lamp to achieve full brightness before the snap. It is only used when the Lamp DO is non zero and the Delay value is non zero.

Enable (Logging). This flag enables logging to the TP default device and path. The filename is iRTorchMate.CSV. When the box has an x, it is enabled. It logs the found user tool or the reason for a failure. The user is responsible for file maintenance. The IRTorchMate.csv file will grow until it is out of memory and logging will be lost, and error may be posted.

Log Count is the log number. It is incremented for each logging. The log number is one of the fields in the log file. Setting this value to 0 will cause headers to be logged to the log file and restarts the count.

App Frame No is the application user frame number.

Temp Frame No is the user frame number for a temporary frame.

View Tool No is the tool number of the view tool.

Grid Tool No is the grid tool number.

8.7. Status Screen

The STATUS screen is used to indicate the status of iRVision TorchMate and to recover from errors. The screen fields are color coded to highlight areas of concern. Field back ground colors meanings are defined in the following table.

<u>Background Color</u>	<u>Meaning</u>
Orange	<u>ABORT</u>
Red	<u>ERROR</u>
Yellow	<u>WARNING</u>
CYAN	<u>MESSAGE</u>
White	<u>NORMAL</u>
Gray	<u>NOT USED</u>

Figure 8-9 Background Color Table

The Status screen in Figure 8-10 displays no errors

Status

iRTorchMate Status

Torch Index 1 Robot Group 1

Errors: 0 Warnings: 0 Repeats: 0

Success

Wire 1 Status: 0 Wire 2 Status: 0

X: 0 W: 0 X: 0 W: 0

Y: 0 P: 0 Y: 0 P: 0

Z: 0 R: 0 Z: 0 R: 0

[TYPE] ***** SETUP STATUS

Figure 8-10 Status Screen

If the status fields have large value (greater than +/- 5), then the nominal value has never been saved. The displayed values reflect the difference between the nominal “golden tool” and the current tool. Warnings or Errors area logged depending on the difference of the change.

Torch Index. Displayed data is for a specific torch. This field allows the user to select a torch. It must be a valid torch number.

Robot Group is the group number of the robot holding this torch.

Warnings. This field displays the number of warnings issued since the warning count was set to zero. Reset the warning count using the SETUP Threshold screen.

Errors. This field displays the number of errors issued since the error count was set to zero. Reset the warning count using the SETUP Threshold screen.

Repeats. This field displays the number of consecutive warnings since the last error or success. After the maximum number of repeats is reached, an excessive warnings error is posted.

Wire 1 Status. This field displays the status of wire 1. If any of its elements (XYZWPR) exceed their warning limit or error limit, a non-zero value appears. A ‘1’ indicates a warning. A ‘2’ indicates an error. A ‘3’ indicates both a warning and an error.

Wire 2 Status. This field displays the status of wire 2. If any of its elements (XYZWPR) exceed their warning limit or error limit, a non-zero value appears. A ‘1’ indicates a warning. A ‘2’ indicates an error. A ‘3’ indicates both a warning and an error.

X. This field displays the absolute value of the difference of the current X value and the nominal X value.
X value. X. This field displays the absolute value of the difference of the current X value and the nominal X value.

Y. This field displays the absolute value of the difference of the current Y value and the nominal Y value.

Z. This field displays the absolute value of the difference of the current Z value and the nominal Z value.

W. This field displays the absolute value of the difference of the current W value and the nominal W value.

P. This field displays the absolute value of the difference of the current P value and the nominal P value.

R. This field displays the absolute value of the difference of the current R value and the nominal R value.

8.8. Model Status Screen

The Model Status screen displays the XYZWPR location of the found model in Tool 0 coordinates (face plate). There are three models per wire. A two wire torch has six models. The first model is the location of the bottom of the wire and projected onto the center line. The second model is the lowest point of the wire (very close to model 1). The third model is the high point on the wire in the Search Area.

The screenshot shows a screen titled "Status: Model User Tool 0". Below the title, it says "iRTorchMate Model User (Tool 0)". The main display area shows the following information:

```

Model Index: 1                      Measured LL Distance: 2.6429e-005

Model Pos

CONF:N D B, 0, 0, 0
X: -198.576  Y: 165.267  Z: 179.281
W: 2.484    P: -1.993   R: -0.086
  
```

At the bottom of the screen, there is a navigation bar with buttons labeled "[TYPE]", "SETUP", "STATUS", and a partially visible button on the right.

Figure 8-11 Model Status Screen

Model Index. This field indicates the model number. There are three models per wire.

Model 1. Bottom wire tip location of wire 1. (origin)

Model 2. Minimum point location on wire 1

Model 3 Maximum point location on wire 1

Model 4. Bottom wire tip location of wire 2. . (origin)

Model 5. Minimum point location on wire 2

Model 6 Maximum point location on wire 2

Measured LL Distance. This field measures the error used when combining the intersection of the two camera views. Ideally the two camera lines intersect exactly. Of course there is always some error the smaller the better. The measurement for model 1 and 4 are calculated differently then models 2, 3, 5, and 6. All the points for each wire are used to calculate the best line fit. The

distance is how far the line connecting the camera for view 1 and the model are from the final wire. Model 1 and 4 is the distance of the 3D tip point and the wire. The 3D tip point is the intersection of the tip with view 1 camera and view 2.

Model Pos is the XYZ location with respect to the face plate.

8.9. Current Tool Screen

This shows the value of the current tool.

```

Status: Current UTOOL
iRTorchMate Current UTOOL

Torch Index: 1      Wire Index: 1      UTOOL[ 1 ]

Current Tool

CONF: N D B, 0, 0, 0
X: -198.576 Y: 165.267 Z: 179.281
W: -179.766 P: -5.746 R: 47.980

[ TYPE ]  SETUP  STATUS

```

Figure 8-12 Current Tool Screen

Torch Index is used to change the torch number

Wire Index is used to change the wire number. Wire 1 is right side. Wire 2 is left side.

UTOOL is the MNUTOOL assigned to this wire

8.10. Nominal Tool Screen

This shows the value of the nominal tool. The nominal tool is used to compare changes with the current tool.

Status: Nominal UTOOL

iRTorchMate *Nominal UTOOL*

Torch Index: 1 Wire Index: 1 UTOOL[1]

Nominal Tool

CONF:N D B, 0, 0, 0

X: -198.576 Y: 165.267 Z: 179.281

W: -179.766 P: -5.746 R: 47.980

[TYPE] SETUP STATUS

Figure 8-13 Nominal Tool Screen

Torch Index is used to change the torch number

Wire Index is used to change the wire number. Wire 1 is right side. Wire 2 is left side.

UTOOL is the MNUTOOL assigned to this wire

8.11. Tool Offset Screen

This shows the value of the tool offset. The tool offset is the value to calculate the new tool. The \$MNUTOOL[x] is set to this value.


```

Status: Tool Offset
iRTorchMate Tool Offset

Torch Index: 1      Wire Index: 1      UTOOL[ 1 ]

TOOL OFFSET

CONF: N D B, 0, 0, 0
X: 0.000 Y: 0.000 Z: -0.000
W: 0.000 P: 0.000 R: -0.000

[ TYPE ]  SETUP  STATUS

```

Figure 8-14 Tool Offset Screen

Torch Index is used to change the torch number

Wire Index is used to change the wire number. Wire 1 is right side. Wire 2 is left side.

UTOOL is the MNUTOOL assigned to this wire

8.12. Inverse Tool Offset Screen

This shows the value of the tool offset that is applied in the MOVE instruction that contains a TOOL OFFSET clause. It is the inverse of the Tool Offset value.

Status: Inverse UTOOL Offset

iRTorchMate Inverse UTOOL Offset

Torch Index: 1 Wire Index: 1 UTOOL[1]

Inverse Tool Offset

CONF: N D B, 0, 0, 0

X: -0.000 Y: -0.000 Z: 0.000

W: 0.000 P: 0.000 R: -0.000

[TYPE] SETUP STATUS

Figure 8-15 Inverse Tool Offset Screen

Torch Index is used to change the torch number

Wire Index is used to change the wire number. Wire 1 is right side. Wire 2 is left side.

UTOOL is the MNUTOOL assigned to this wire.

9

SAVING AND RESTORING

9. SAVING AND RESTORING

9.1. TP Programs

Teach Pendant programs can be saved using the normal method found in the operator's manual. The list of TP programs used with iRVision TorchMate is listed below.

These programs are examples that the user can copy for their application.

- itrwire.tp
- irtsetframe.tp
- irtcalib.tp
- irtUtility.tp
- irtShow.tp

9.2. KAREL Data

All KAREL data is stored in the KAREL program, IRTDATA. The VR data can be saved to unique file names. Use KCL to save a VR file

Syntax: SAVE VARS < **file_name** > <= **prog_name** >

Example: SAVE VARS irtdata1 = irtdata

See the **KAREL Reference Manual's** Appendix – *KCL Command Alphabetical Description* for complete details. The user can also write a KAREL program to perform the KCL command.

9.3. Vision Data

Vision data files are stored on the *FR:* device. The file can be shown on the Teach Pendant using the Menu Files command. Set the device to *FR:* and use the directory command for *.vd.

Copy the file to an MC: or UD1 device.

10

LOGGING

10. LOGGING

10.1. Overview

The logging is intended as a temporary analysis tool. There is no reason to leave the logging enabled for a long period of time. Logging allows the user to save User Tool results to a csv type file that is Excel friendly. To view the file with EXCEL, copy the *iRVision TorchMate.csv* to a PC that supports EXCEL.

10.2. Activating the Logging

Use logging after placing a memory card in the MC: device. Select the device and folder with the TP's File Menu. Logging can be enabled/disabled with the **Logging** check box on the *Flags Setup* screen. When logging is enabled data is saved to the file path defined by the current TP file menu.

10.3. File Maintenance

The log file name is *iRVision TorchMate.csv*. Normally the data path is assigned to the MC: device.

The user is responsible for ensuring that the device does not become full. Delete or move the file to another location when it gets large. If the file does become full, a stream of annoying messages will appear on the TP status line. If the TP menu default device is assigned to a robot memory device and it becomes full, then this may cause unpleasant consequences.

10.4. iRVision TorchMate.csv Log Data

The information written to the log file are the values of the new Utools. Both tools for a dual wire and the one tool for a single wire are logged. The file headers for the CSV file are:

Count
Time Data
1stTool Data
Tool No
X
Y
Z
W
P
R
2ndToolData
Tool No
X
Y
Z
W
P
R
Result Code
Result (description of an error or Success)

A new row of data is added to the CSV file after each robot visit to the Inspection Box. If the inspection fails, no position data is included in the logging. However, the Count, Date and Time, Result Code and Result are still logged. Error results for both the *Snap and Find 1* and the *Snap and Find 2* functions are logged

If the inspection is successful, the *Snap and Find 2* results are logged. There are no successful results logged for the *Snap and Find 1* function.

10.5. Resetting the Log Count

To send a new header to the CSV file, set the **Log Count** value to 0. The next result logged will write the header to one row and the results to the next roll. The **Log Count** can be reset to 0 using the *Flags Setup* screen.

A

ERROR CODES

A ERROR CODES

This section describes the error codes, severity, and description. Warnings are logged to the Alarm log. Errors (PAUSE) are logged and require operator intervention to recovery. ABORT causes the program to halt. The problem must be fixed before continuing.

Some of the error messages list secondary data shown in **green**.

Error Code	Severity	Description
<i>IRTSuccess</i>	NONE	<i>No Errors or warnings</i>
irt_e_FOCPTN	ABORT	<i>Invalid camera to target dustance < 100mm</i>
irt_e_NOTLIN	WARN	<i>Not Linear</i>
irt_e_MODELi1	WARN	<i>GET OFFSET wrong model ID model ID</i>
irt_e_FINDFA	WARN	<i>RUN FIND failed reason</i>
irt_e_IVDTHO	ABORT	<i>Invalid number of Wires(tools)</i>
irt_e_PARAL1	WARN	<i>Wire Wire #:Two views are parallel</i>
irt_e_BADBAS	ABORT	<i>Tool model ID End or Base cannot be 0</i>
irt_e_WTHRES	WARN	<i>Element exceeds Warning threshold</i>
irt_e_ETHRES	PAUSE	<i>Element exceeds Error threshold</i>
Irt_e_INVALID	Warn	<i>Invalid: Device or index</i>
irt_e_PARAL1	ABORT	<i>Get Focal Failed reason</i>
irt_e_SAMEPT	WARN	<i>Model model ID and Model Model ID are same point</i>
irt_e_P3SR	ABORT	<i>SET_INT_REG failed</i>
IRT_E_CONSEC	PAUSE	<i>Number of consecutive warnings exceeds threshold</i>
IRT_E_OPRRSP	WARN	<i>Operator Response was (ABORT, IGNORE,RETRY)</i>

Table A-1 Error Codes

B

PROGRAMMING AND CUSTOMIZING IRVISION TORCHMATE INSPECTION

B PROGRAMMING AND CUSTOMIZING IRVISION TORCHMATE INSPECTION

The IRT software supports multiple functions. The KAREL program irtMain can be called by a Teach Pendant program or another KAREL program. This is the runtime interface. The user can write programs to perform vision tests, set nominal data, clear trend data, and query test settings. The table below displays the supported functions

B.1 Calling a Function

Call IRTMAIN using

CALL iRTmain(FC, StringIn, Status Index, PR index, String Index, R index)

FC	-- Function Code
StringIn	-- String constant
Status Index	-- Register to receive status (0 is success)
PR index	-- Index of Position Register
String index	-- Index of String Register
R index	-- Index of General Register

B.1.1 KAREL Call

This method is the easiest to implement, but it requires knowing the function codes. The first parameter of the KAREL call is always the function code. The remaining arguments are the same.

B.1.2 TP Call

This method has the name of the function encrypted in the name of the TP program. It has one less argument, the function code is missing. The advantage of this method is the comments in the TP program describe the arguments and make it easier to remember. The only disadvantage is requiring an extra program. The user could have used the KAREL call method and then the TP program would not need to be on the robot. Since all the TP programs are MACRO type they do not appear in the Select command unless the Type is set to all or Macro.

B.2 Function Description and Argument List

The table below indicates which arguments are valid for a function.

Name (String function)	Function Value	Vis Proc	Status Reg	PR Reg	String Index	Register Index
Inquire	20	'NTools' / GroupNo	O	-	-	#tools / Group#
RunSnap1	1	R	O	-	-	-
RunSnap2	2	R	O	-	-	-
GetTool	4	-	R	O	-	-
SetTorch	6	-	O	-	-	R
SetVPName	8	R	O	O	-	-
Set Group	11	-	O	-	-	R
SetNom	12	-	O	-	-	R
Logging	14	-	O	-	-	1/0

Table B-1 IRTMAIN Functions

O indicates optional, R indicates Required, - indicates not used.

Note in many cases the Register Index is a direct value see examples

Parameter List

The parameters are used differently for each function. See the *IRTUTILITY.TP* program for examples of how the functions are used.

B.3 IRTMAIN Functions

B.3.1 Inquire

This function requests IRVision TorchMate to return a value about the current setup. The user can request the number of wires in the current torch or the robot group number of the current torch. The value is returned

Example:

```
R[15:status] = 0
```

```
CALL iRTmain('Inquire' 'NTools', 15,0,0,6 )
```

'Inquire' is the function name.

'Ntools' or 'GroupNo' are the valid inquiry types

R[15] is zero if successful

R[6] will contain a 1 or 2 for an *Ntools* request or a 1through 4 for group number.

B.3.2 Logging

This function requests IRVision TorchMate to enable or disable logging. The user can request the number of wires in the current torch or the robot group number of the current torch. The value is returned

Example:

```
R[15:status] = 0
```

```
CALL iRTmain('Logging' 0, 15,0,0,1/0 )
```

'Logging' is the function name.

'Ntools' or 'GroupNo' are the valid inquiry types

R[15] is zero if successful

The sixth argument is a 1 to enable or 0 (or missing) to disable logging.

B.3.3 RunSnap1

This function requests IRVision TorchMate to Snap an image while at the view 1 position. The robot must be at the View 1 position (-45 degrees). All the points along the wire(s) are found.

Example:

```
R[15:status] = 0  
  
CALL iRTmain('RUNSNAP2' 'IRTWIRE', 15 )  
  
R[15] is zero if successful  
  
IRTWIRE is the vision process name  
  
RUNSNAP1 is the function code
```

B.3.4 RunSnap2

This function requests IRVision TorchMate to snap an image view 2 position. The robot must be at the View 2 position (+45 degrees). All the points along the wire(s) are found. The tool offset is calculated and the UTOOL is updated.

Example:

```
R[15:status] = 0  
  
CALL iRTmain('RUNSNAP2' 'IRTWIRE', 15 )  
  
R[15] is zero if successful  
  
IRTWIRE is the vision process name  
  
RUNSNAP2 is the function code
```

B.3.5 GetTool

This function requests IRVision TorchMate to calculate the User Tool . The Runs Snap 1 and Run snap 2 functions must be performed before calling this function.

Example:

```
CALL iRTmain('GETTOOL',0,15)  
  
GETTOOL is the function code  
  
R[15] is an optional status register
```

B.3.6 Init

This function requests IRVision TorchMate to clear all data and reset to the factory defaults.

Example:

```
CALL iRTmain('INIT')
```

INIT is the function code

B.3.7 SetTorch

This function requests IRVision TorchMate to change the torch number. This is required in multi torch system.

Example:

```
CALL iRTmain('SETTORCH',0,0,0,0,2)
```

SETTORCH is the function code

2 is the new torch number

B.3.8 SetVPName

This function requests IRVision TorchMate to change the Vision Process Name for the current gun data. This is a fast way to change the VP name . The user could use the View Setup screen instead.

Example:

```
CALL iRTmain('SETVPNAME','IRTWIRE1',4,0,0,1)
```

```
CALL iRTmain('SETVPNAME','IRTWIRE2',4,0,0,2)
```

SETVPNAME is the function code

IRTWIRE1 and IRTWIRE2 are the new vision process names

R[4] is the optional status register. It is 0 if successful

1 and 2 are the view numbers

B.3.9 SetGroup

This function requests IRVision TorchMate to change the robot group number. This is used in multi gun system. Note by default the group number is retrieved from the GUN Setup screen. This function is intended to be used with the INQUIRE function.

Note: the SetTorch will set the group number after the torch index has been updated. Using the Torch Setup screen provide the same function. This provides a method to edit the Group number from program control.

Example:

```
CALL iRTmain('SETGROUP',0,0,0,0,2)
```

SETGROUP is the function code

2 is the new group number for the current gun

B.3.10 SetNom

This function requests IRVision TorchMate to set the nominal flag. The next time a UTOOL is updated (RUNSNAP2) the Nominal UTOOL is set to the new UTOOL. All threshold tests are performed comparing the nominal and current tool. This function can also be performed using the Control Flags screen. Note the flag is normally cleared after a tool has been measured, the nominal tool is set to the measured tool, and the flag is cleared.

Example:

```
CALL iRTmain('SETNOM',0,0,0,0,2)
```

SETGROUP is the function code

2 is the new group number for the current gun

C

USING IRTSETUP_T1.TP

C USING IRTSETUP_T1

The user should use Appendix C for Versions 9.10P/10 and later. For earlier versions the user should use Appendix D.

This is what the user will see when running the IRTSETUP_T1.TP program in step mode. Comments have been added to explain what the user should do. Section name have been added to make navigating to the desired section easier.

The program has four paths that are defined by the number of wires on a torch (1 or 2) and the method selected for training the initial torch wire (one Point method or the six point manual method found in MENU>SETUP>FRAMES).

The number of wires is defined in the IRTorchMate Torch Setup screen using the Number of Wires field. The Utool training method is defined in the iRTorchMate Control Flags screen using the Use 1 Point flag.

The program performs 21 steps for setting up Torch Mate. Many of the steps are automatic. Some steps have a PAUSE instruction to allow the user to perform the step.

The text changes color when the program changes. For example, when IRTSETUPT1 calls IRTCALIp_T1 the text changes to green.

C.2 Set Torch Number

The Torch data area allows for four torches. The default is one. It is set in line 6.

```
1:  !----- ;
2:  !iRTorch Mate SETUP Procedure ;
3:  !----- ;
4:  ;
5:  ! SET TORCH NO ;
6:  CALL IRTMAIN('SetTorch',0,0,0,0,1) ;
7:  ;
```

C.3 Step 1. Install Inspection Box

Place the inspection box in a locations that can be easily reached by the robot. For best results and easiest setup place the box where the robot in the box position has the X axis of the UTOOL pointed toward the camera and is able to rotate plus or minus 45 degrees about the Z axis of the UTOOL.

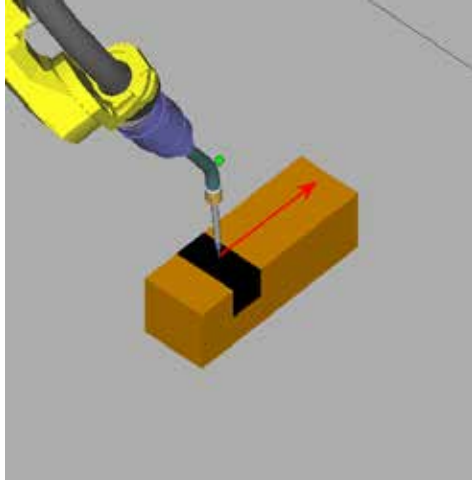


Figure 10-1 X-axis of UTOOL in box

```
8:  !1)Install firmly the iRVision ;
9:  !Inspection Box in the Robot Cell ;
10:  ;
```

C.4 Step 2. Connect Camera Cable

```
11:  !2)Connect Camera Cable ;
12:  ;
```

C.5 Step 3. Connect LED Light Cable

```
13:  !3)Connect LED Light Cable ;
14:  ;
```

C.6 Initialize all KAREL data

```
15:  !Initialize all KAREL data ;
16:  CALL IRTUTILITY_T1(10) ;
17:  ;
```

C.7 Step 4. Setup iRTorchMate parameters

```
18:  !4)Setup iRTorch Mate parameters ;
19:  !in SETUP Screens ;
20:  ;
```

C.8 Step 5. Train the Master UTOOLS

```
21: !5) Train the Master UT00Ls ;
22: ;
```

C.9 Step 6. Relocate the Camera in iRVision

The next step is to teach the calibration grid paddle as a user tool. The camera needs to be relocated to a friendlier position to allow the robot large angle of motion while in the camera field of view. The user can just as easily mount the camera on a tripod.

```
23: !6)Remove the Camera in iRvision ;
24: !Inspection Box,turn it by 90deg ;
25: !and fix it again (sideways) ;
26: ;
```

C.10 Step 7. Setup Camera (IRTCALIB_T1)

The user should enter the vision setup page for camera data setup. If your controller only has one camera and it agrees with the **Port Number** and **Camera Type**, this step can be skipped.

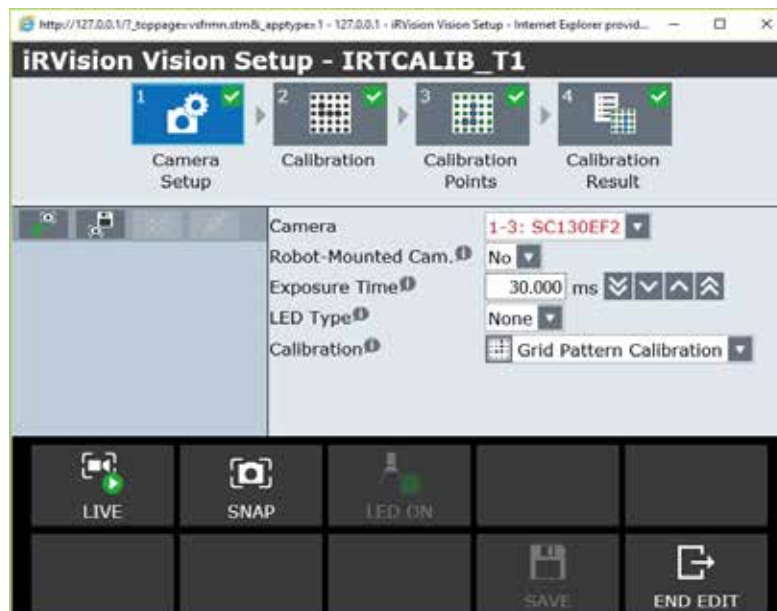


Figure 10-2 Default Camera Setup

```
27: !7)Setup Camera (IRTCALIB_T1) ;
28: ;
```

C.11 Step 8. Install Calibration Grid

Attach the paddle grid to the torch. The large corner circle on the grid will be taught as the UTOOL center point. The Tool X axis will align with the long side of the letter 'L' and the y axis will align with the short side of the letter 'L'. The letter 'L' refers to the pattern formed by the four large circles on the grid.

```
29:  !8)Install Calibration Grid on ;
30:  !Torch's neck and place it in FOV ;
31:  ;
```

C.12 Step 9. Teach UTOOL[7]

Use the IRVISION utility Automatic grid frame set to teach the tool. The UTOOL should be set to 7. When this operation completes UTOOL[7] will be the grid paddle tool.

```
32:  !9)Teach UTOOL[7] using ;
33:  !MENU>IRVISION>IRVISION UTILITIES ;
34:  !AUTOMATIC GRID FRAME SET ;
35:  !then verify it with crosshair ;
36:  ;
```

Once UTOOL[7] (the paddle grid) has been verified it is safe to return the camera to the inspection box. Do not remove or bump the paddle grid.

Note: if you have two robot groups sharing the same inspection box (same camera), a second paddle should be placed on robot of group2 and the IRTSETUP_T2 should be run to this point. Once the camera is calibrated it can not be moved from the inspection box. A second method would be to use a second camera just for performing the AUTOMATIC GRID FRAME SET.

C.13 Step 10. Remove the Camera

```
37:  !10)Remove the Camera and ;
38:  !reinstall it in the original pos ;
39:  ;
```

C.14 Step 11. Check and re-adjust exposure

```
40:  !11)Check and re-adjust exposure ;
41:  !time to correct values for ;
42:  !original Camera position (5ms) ;
43:  ;
```

C.15 Step 12. Teach UFRAME of iRvision Box

Jog the robot into the opening of the inspection box with the paddle grid facing the camera. It should be centered in the opening. Using LIVE mode and the crosshairs of the camera center and align the grid on the cross hairs. Set the distance of the grid to camera to be the middle of the inspection opening. The grid will be taught as the application frame of the inspection box. The IRTSETFRM program will simplify teaching the frame. The procedure differs slightly from the standard Menu method (Uses origin, x axis, and y axis). This procedure uses origin , X axis and Z axis.

```

44:  !12)Teach UFRAME of iRvision Box ;
45:  !by using IRTSETFRM_T1.TP ;
46:  !Execute in STEP Lines 1,2 and 3 ;
47:  !to enable correct UT and UF ;
48:  !Then touchup P[1],P[2] and P[3] ;
49:  ;
50:  CALL IRTSETFRM_T1      ;
51:  ;

```

C.16 Step 12.1 IRTSETFRM.TP**IRTSETFRM.TP**

```

1:  !GRID UT00L ;
2:  CALL IRTMAIN('SetTool','Grid') ;
3:  CALL IRTMAIN('FrameZero') ;
4:  ;

```

Move the cursor to line 6 and teach the three points for the frame. The user should jog in the Tool frame. X will be up and Z toward the camera relative to a LIVE camera. Expect the Z position to need adjustment. After recording the three positions cursor to line 11 and continue in step mode.

```

5:  JMP LBL[10] ;
6:L P[1:ORIGIN] 500mm/sec FINE      ;
7:L P[2:X DIR] 100mm/sec FINE      ;
8:L P[3:Z DIR] 100mm/sec FINE      ;
9:  END ;
10: ;
11: LBL[10] ;
12: !Save PR[77] in PRTables{-77} ;
13: CALL IRTMATH('MovePR',0,0,(-77),77) ;
14: PR[77]=P[1:ORIGIN] ;
15: CALL IRTMATH('MovePR',0,0,(-51),77) ;
16: PR[77]=P[2:X DIR] ;
17: CALL IRTMATH('MovePR',0,0,(-52),77) ;
18: PR[77]=P[3:Z DIR] ;
19: CALL IRTMATH('MovePR',0,0,(-53),77) ;
20: !PR[77] = Calculated UFRAME ;
21: CALL IRTMATH('FrameOXZ',0,0,77,(-51),(-52),(-53)) ;
22: CALL IRTMAIN('SetFrame','App',0,77) ;

```

The user frame has been taught. P[4] is already taught as (0,0,0,0,0,0). It may have the wrong configuration and require a touch up. If touch up is required manually set XYZWPR to zeroes and move to it. The robot should be centered in the user frame.

FYI. This is a useful tip for debugging. The robot position is (0,0,0,0,0,0) and the LIVE cross hair is at (0,0,0,0,0,0). Whenever you move the torch to (0,0,0,w,p,r) the wire tip will be on the LIVE crosshair. Note the WPR may not be 0,0,0 for other UTOOLS.

```
23:L P[4] 100mm/sec FINE      ;
24: !Restore PR[77] ;
25: CALL IRTMATH('MovePR',0,0,77,(-77)) ;
```

C.17 Step 13 Setup IRTCALIB_T1 in Vision

Now that the application frame has been taught, the camera can be calibrated. Verify that the IRTCALIB_T1 vision date (Camera Calibration Setup) is using Application frame 1 as the user frame and UTOOL 7 as the user tool. If the user is using a different Application frame or Grid tool, these values will need to be edited. The exposure time may need to be adjusted if the grid is too bright or dark. Exit calibration after making modifications.

```
52: !13)Setup IRTCALIB_T1 in Vision ;
53: !Setup PC Screen then you MUST ;
54: !close Calibration Setup screen! ;
55: ;
56: CALL IRTCALIB_T1      ;
57: ;
```

C.18 Step 13.1 IRTCALIB_T1.TP

The user may need to TouchUp P[1,] P[2], and P[3] if the configuration changes. Normally the Positions are correct.

IRTCALIB_T1

```
1: !INSPECTION FRAME ;
2: CALL IRTMAIN('SetFrame','App') ;
3: !GRID UTOOL ;
4: CALL IRTMAIN('SetTool','Grid') ;
```

P[1] and P[2] may need to be touched up if the configuration differs.

```
5:L P[1:ORIGIN] 100mm/sec FINE      ;
6: ;
7:L P[2:PLANE 1] 100mm/sec FINE      ;
8: CALL IRTMAIN('CamCalib1','IRTCALIB_T1') ;
9:L P[3:PLANE 2] 100mm/sec FINE      ;
```

```
10: CALL IRTMAIN('CamCalib2','IRTCALIB_T1') ;  
11:L P[1:ORIGIN] 100mm/sec FINE ;
```

C.19 Step 14. Remove grid from torch neck

The application user frame teaching and camera calibration setup have been completed. Remove the grid paddle.

```
58: !14)Remove grid from torch neck ;  
59: ;
```

C.20 Step 15. Train UTOOL[4] using six point method

The user should train the UTOOL using the Single UTOOL number (default is 4). The TCP location does not need to be too accurate since one of the later steps will fine tune the Single UTOOL TCP. The user should concentrate on getting the Axis directions correct. Use Menu>Setup>Frames>Tools. Please note that the UTOOL[4] must be taught at any stage before this step. Please refer to Section C.19 in case of tandem TCP.

```
60: !15)Jog Robot to have Wire tip ;  
61: !in the center of LIVE crosshair ;  
62: ;  
63: PAUSE ;
```

C.21 Train UTOOL[1 & 2] using six point method in case of a Tandem TCP

Please note the direction of the axis as shown in the figure below while you are teaching the UTOOL 1 and UTOOL 2.

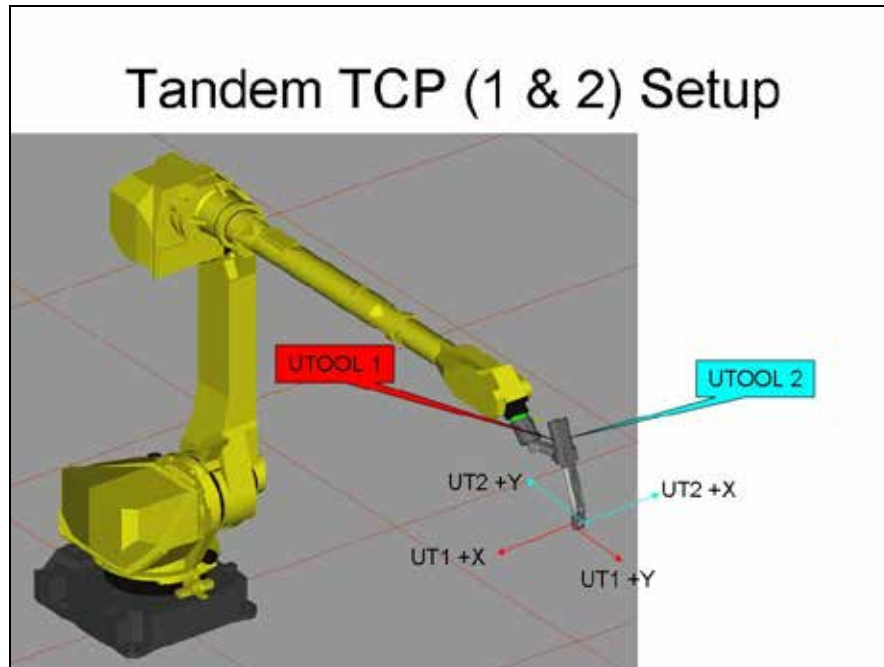


Figure 10-3 Position of Torch (DUAL) in box

Please run IRTUTILITY_T1[93] for updating the View Tool. The program snippet pauses at line 130. The user should Touch Up P[4](Line 132). Note: the position of the robot in the inspection Box differs by 90 degrees (roll) when compared with a single wire torch. The view position is taught between the two wires.

```

126: LBL[93:SET VIEW] ;
127: CALL IRTMATH('MovePR',0,0,(-77),77) ;
128: CALL IRTMAIN('SetFrame','App') ;
129: CALL IRTMAIN('ToolZero') ;
130: PAUSE ;
131: !Train Wire with LIVE cross hairs ;
132:L P[4] 100mm/sec CNT100 ;
133: PR[77]=LPOS ;
134: CALL IRTMATH('Inverse',0,0,77,77) ;
135: CALL IRTMAIN('SetTool','View',0,77) ;
136: CALL IRTMATH('MovePR',0,0,77,(-77)) ;
137: END ;

```

C.22Step 16. Train Vision Process

Please refer to Chapter 7.

```

64: !16)Train IRTWIRE_T1 Vision Proc ;
65: !inside IRT_T1 ;

```

```

66:  CALL IRT_T1      ;
67:      ;
68:  PAUSE ;

```

C.23 Step 17. Get Number of wires

This step is automatic. It performs an inquiry of the data base to determine if this torch has one wire (SINGLE torch) or two wires (DUAL torch). Edit the **Number of Wires** field in the Torch Setup Screen to define the number of wires used with the torch.

Setup: Torch

iRTorchMate Torch Setup

Torch Index 1 Group No 1

Number of Wires 2 P2W Threshold 5

Local View Tool [0]

☐ Multi Pass ☐ Calc Fit Error

Figure 10-4 Number of Wires in Torch Setup Screen

```

69:  !17) Update Master tools ; - probably, step 18
70:  CALL IRTMAIN('INQUIRE','NTOOLS',0,0,0,77) ;
71:  IF R[77]=1,JMP LBL[2010] ;
72:  IF R[77]=2,JMP LBL[2020] ;
73:  PAUSE ;
74:  ABORT ;
75:      ;

```

C.24 Step 18. Set Master Tool to Single UTOOL

This step is automatic and will copy UTOOL[4] to UTOOL[8].

```

76:  LBL[2010] ;
77:  !SET MASTERS TO SINGLE ;
78:  CALL IRTUTILITY_T1(140) ;
79:      ;

```

C.25 Step 18.1 IRTUTILITY_T1(140)

```

208:  LBL[140:SET UTOOL 8 TO 4] ;
209:  CALL IRTMATH('MovePR',0,0,(-77),77) ;
210:  CALL IRTMAIN('SaveTool','Single',0,77) ;

```



```

211: CALL IRTMAIN('SetTool','MasterS',0,77) ;
212: CALL IRTMATH('MovePR',0,0,77,(-77)) ;
213: END ;
214: ;

```

C.26 Manually Override Master UTOOL

The user has the option to make the Master UTOOL WPR values perfect based on expected values. The WPR should match expect values from a perfect UTOOL (based on drawings).

SETUP Frames			^ i
Tool Frame	Direct Entry		4/7
Frame Number: 8			
1	Comment:	Eoat8	
2	X:	-1045.8>	
3	Y:	-628.495	
4	Z:	432.402	
5	W:	179.952	
6	P:	-28.625	
7	R:	1.084	
Configuration:		N U T, 0, 0, 0	

Figure 10-5 Example of Master UTOOL values before adjustment

The user knows the WPR of a perfect UTOOL shown in C-4 should be 180, -30, and 0 for WPR. Use the direct entry method to modify the WPR values as shown in the figure below.

SETUP Frames			^ i
Tool Frame	Direct Entry		4/7
Frame Number: 8			
1	Comment:	Eoat8	
2	X:	-1045.8>	
3	Y:	-628.495	
4	Z:	432.402	
5	W:	180.000	
6	P:	-30.000	
7	R:	0.000	
Configuration:		N U T, 0, 0, 0	

Figure 10-6 Example of Master UTOOL values after adjustment

```

80: !MANUALLY OVERRIDE MASTER ;
81: JMP LBL[2030] ;
82:

```

C.27 Step 19. Set the Master UTOOLS[9 & 10] to the Dual UTOOLS[1 & 2]

This is an automatic step that copies UTOOL[1] to UTOOL[9] and copies UTOOL[2] to UTOOL[10].

```

83:  LBL[2020] ;
84:  !SET MASTERS TO DUAL ;
85:  ;
86:  CALL IRTUTILITY_T1(120) ;
87:  CALL IRTUTILITY_T1(160) ;
88:  ;

```

C.28 Step 20. Manually Override the Master UTOOLS

This step is performed manually. See the Previous Step 20. Repeat the procedure for both UTOOL[9] and UTOOL[10].

```

89:  !MANUALLY OVERRIDE MASTER ;
90:  LBL[2030] ;
91:  ;

```

C.29 Set Nominal Flag

```

92:  !Set Nominal Flag ;
93:  !now that tool is nominal ;
94:  !and re-run IRT_T1 ;
95:  CALL IRTUTILITY_T1(30) ;
96:  ;
97:  CALL IRT_T1      ;
98:  ;

```

C.30 IRT_T1.TP

```

4:  LBL[10:Loop] ;
5:  ;
6:  !INSPECTION FRAME ;
7:  CALL IRTMAIN('SetFrame','App') ;
8:  !View UTOOL ;
9:  CALL IRTMAIN('SetTool','View') ;

```

J[1] is the entrance/exit position of box, P[2] is view 1 position, and P[3] is view 2 position. None of these positions should need to be Touched Up.

```

10: J P[1:OVER BOX] 100% FINE      ;
11: L P[2:VIEW -45 DEG] 200mm/sec FINE      ;
12: !Run Snap View 1 R[1] is status ;

```

RUNSNAP1 finds the points of the wire at the view 1 position. Watch the runtime screen to verify that the wire is in the search window. Adjust the search window and repeat Line 13 if wire is not in search window.

```
13: CALL IRTMAIN('RUNSNAP1',0,1) ;
14: IF R[1]<>0,JMP LBL[100] ;
15:L P[3:VIEW 45 DEG] 2000mm/sec FINE ;
```

RUNSNAP2 finds the points of the wire at the view 2 position. Watch the runtime screen to verify that the wire is in the search window. Adjust the search window and repeat from Line 11 if wire is not in search window. (This will repeat RUNSNAP1 also).

```
16: !Run Snap View 2 R[1] is status ;
17: CALL IRTMAIN('RUNSNAP2',0,1) ;
18: IF R[1]<>0,JMP LBL[100] ;
```

The robot exits the box to P[1] and calls the GETTOOL function which defines the UTOOL(s). R[1] in Line 21 is defined as the status register.

```
19:L P[1:OVER BOX] 500mm/sec FINE ;
20: !GET TOOL R[1] is status ;
```

GETTOOL function calculates the User Tool

```
21: CALL IRTMAIN('GETTOOL',0,1) ;
```

Jump to different labels for each outcome

```
22: LBL[100] ;
23: SELECT R[1]=0,JMP LBL[500] ;
24:     =1,JMP LBL[501] ;
25:     =2,JMP LBL[502] ;
26:     =3,JMP LBL[503] ;
27:     =4,JMP LBL[504] ;
28:     =5,JMP LBL[505] ;
29:     ELSE,JMP LBL[510] ;
30: ;
31: LBL[500:Success] ;
32: LBL[502:Ignore] ;
33:L P[1:OVER BOX] 500mm/sec FINE ;
34: END ;
35: ;
36: LBL[503:Retry] ;
37: LBL[505:Retest] ;
38:L P[1:OVER BOX] 100mm/sec FINE ;
39: JMP LBL[10] ;
40: ;
41: LBL[501:Pending] ;
42: LBL[510:Invalid] ;
43:L P[1:OVER BOX] 500mm/sec FINE ;
```

```
44:  ABORT ;
45:  END ;
46:  ;
47:  LBL[504:Abort] ;
48:L P[1:OVER BOX] 500mm/sec FINE ;
49:  ABORT ;
```

```
143:  ;
```

C.31Setup Complete

```
99:  !Nominal UT00L flag should be ;
100:  !cleared automatically. ;
101:  !From next inspection if UT ;
102:  !exceeds thresholds, messages are ;
103:  !posted in iRTorch Mate STATUS ;
104:  ;
105:  !----- SETUP COMPLETED ----- ;
;
```

You are done.

D

USING IRTSETUP_G1.TP

D USING IRTSETUP_G1

Version previous to 8.30P/21 should use Appendix D. Other version of robot software should use Appendix C.

D.2 Introduction

This is what the user will see when running the IRTSETUP_G1.TP program in step mode. Comments have been added to explain what the user should do. Section name have been added to make navigating to the desired section easier.

D.3 Install Hardware

IRTSETUP_G1

```
1:  !-----  
2:  !iRTorch Mate SETUP Procedure  
3:  !-----  
4:  ;
```

See the chapter on hardware for more instruction on how to install the hardware.

```
5:  !1)Install firmly the iRVision ;  
6:  !Inspection Box in the Robot Cell  
7:  ;  
8:  !2)Connect Camera Cable ;  
9:  ;  
10: !3)Connect LED Light Cable ;
```

D.4 Initializing the Database

This call initializes the database to the factor settings.

```
11:  ;  
12: !Initialize all KAREL data ;  
13: CALL IRTUTILITY_G1(10) ;
```

IRTUTILITY.TP

```
7:  LBL[10:INIT DATA] ;  
8:  CALL IRTMAIN('Init') ;  
9:  END ;
```

D.5 Setup the Database

After initializing the data the use should enter the database and setup the cell for their work cell before proceeding. This includes making sure tool assignments are correct and selecting a one wire or two wire torch.

IRTSETUP_G1

```
14:  ;
15:  !4)Setup iRTorch Mate parameters ;
16:  !in SETUP Screens ;
17:  ;
```

D.6 Train Calibration Grid

The next step is to teach the calibration grid paddle as a user tool. The camera needs to be relocated to a friendlier position to allow the robot large angle of motion while in the camera field of view. The user can just as easily mount the camera on a tripod.

```
18:  !5)Remove the Camera in iRvision ;
19:  !Inspection Box,turn it by 90deg ;
20:  !and fix it again (sideways)
21:  ;
```

The user should enter the vision setup page for camera setup. If your controller only has one camera, this step can be skipped.

```
22:  ;
23:  !6)Setup Camera (IRTCALIB_T1) ;
24:  ;
```

Attach the paddle grid to the torch. The large corner circle on the grid will be taught as the UTOOL center point. The Tool X axis will align with the long side of the letter 'L' and the y axis will align with the short side of the letter 'L'. The letter 'L' refers to the pattern formed by the four large circles on the grid.

```
25:  !7)Install Calibration Grid on ;
26:  !Torch's neck and place it in FOV
27:  ;
```

Use the IRVISION utility Automatic grid frame set to teach the tool. The UTOOL should be set to 7. When this operation completes UTOOL[7] will be the grid paddle tool.

```
28:  !8)Teach UTOOL[7] using ;
29:  !MENU>UTILITIES>IRVISION> ;
30:  !AUTOMATIC GRID FRAME SET ;
```

Use the camera setup screen and select LIVE mode with the blue cross hairs displayed. The origin should be under the cross hair. Set the UTOOL to 7 and rotate the robot in TOOL mode. The circle should stay centered on the cross hair.

```
31: !then verify it with crosshair ;
32: ;
```

Once UTOOL[7] (the paddle grid) has been verified it is safe to return the camera to the inspection box. Do not remove or bump the paddle grid.

```
33: !9)Remove the Camera and ;
34: !reinstall it in the original pos
35: ;
36: !10)Check and re-adjust exposure ;
37: !time to correct values for ;
38: !original Camera position (5ms) ;
```

D.7 Train the Application Frame

Jog the robot into the opening of the inspection box with the paddle grid facing the camera. It should be centered in the opening. Using LIVE mode and the crosshairs of the camera center and align the grid on the cross hairs. Set the distance of the grid to camera to be the middle of the inspection opening. The grid will be taught as the application frame of the inspection box. The IRTSETFRM program will simplify teaching the frame. The procedure differs slightly from the standard Menu method (Uses origin, x axis, and y axis). This procedure uses origin , X axis and Z axis.

```
40: !11)Teach UFRAME of iRvision Box ;
41: !by using IRTSETFRM_G1.TP ;
42: !Execute in STEP Lines 1,2 and 3 ;
43: !to enable correct UT and UF ;
44: !Then touchup P[1],P[2] and P[3] ;
45: ;
46: CALL IRTSETFRM_G1 ;
```

IRTSETFRM.TP

```
1: !GRID UTOOL ;
2: CALL IRTMAIN('SetTool','Grid') ;
3: CALL IRTMAIN('FrameZero') ;
4: ;
```

Move the cursor to line 6 and teach the three points for the frame. The user should jog in the Tool frame. X will be up and Z toward the camera relative to a LIVE camera. Expect the Z position to need adjustment. After recording the three positions cursor to line 11 and continue in step mode.

```
5: JMP LBL[10] ;
6:L P[1:ORIGIN] 500mm/sec FINE ;
7:L P[2:X DIR] 100mm/sec FINE ;
8:L P[3:Z DIR] 100mm/sec FINE ;
```



```

9:  END ;
10: ;
11: LBL[10] ;
12: !Save PR[77] in PRTables{-77} ;
13: CALL IRTMATH('MovePR',0,0,(-77),77) ;
14: PR[77]=P[1:ORIGIN] ;
15: CALL IRTMATH('MovePR',0,0,(-51),77) ;
16: PR[77]=P[2:X DIR] ;
17: CALL IRTMATH('MovePR',0,0,(-52),77) ;
18: PR[77]=P[3:Z DIR] ;
19: CALL IRTMATH('MovePR',0,0,(-53),77) ;
20: !PR[77] = Calculated UFRAME ;
21: CALL IRTMATH('FrameOXZ',0,0,77,(-51),(-52),(-53)) ;
22: CALL IRTMAIN('SetFrame','App',0,77) ;

```

The user frame has been taught. P[4] is already taught as (0,0,0,0,0,0). It may have the wrong configuration and require a touch up. If touch up is required manually set XYZWPR to zeroes and move to it. The robot should be centered in the user frame.

FYI. This is a useful tip for debugging. The robot position is (0,0,0,0,0,0) and the LIVE cross hair is at (0,0,0,0,0,0). Whenever you move the torch to (0,0,0,w,p,r) the wire tip will be on the LIVE crosshair. Note the WPR may not be 0,0,0 for other UTOOLS.

```

23:L P[4] 100mm/sec FINE ;
24: !Restore PR[77] ;
25: CALL IRTMATH('MovePR',0,0,77,(-77)) ;

```

D.8 Camera Calibration using IRTCALIB_G1

Now that the application frame has been taught, the camera can be calibrated. Verify that the IRTCALIB_G1 vision date (Camera Calibration Setup) is using Application frame 1 as the user frame and UTOOL 7 as the user tool. If the user is using a different Application frame or Grid tool, these values will need to be edited. The exposure time may need to be adjusted if the grid is too bright or dark. Exit calibration after making modifications.

IRTSETUP_G1

```

47: ;
48: !12)Setup IRTCALIB_G1 in Vision ;
49: !Setup PC Screen then you MUST ;
50: !close Calibration Setup screen! ;
51: ;
52: CALL IRTCALIB_G1 ;

```

IRTCALIB_G1

```

1: !INSPECTION FRAME ;
2: CALL IRTMAIN('SetFrame','App') ;
3: !GRID UTOOL ;
4: CALL IRTMAIN('SetTool','Grid') ;

```

P[1] and P[2] may need to be touched up if the configuration differs.

```

5:L P[1:ORIGIN] 100mm/sec FINE ;
6: ;
7:L P[2:PLANE 1] 100mm/sec FINE ;
8: CALL IRTMAIN('CamCalib1','IRTCALIB_G1') ;
9:L P[3:PLANE 2] 100mm/sec FINE ;
10: CALL IRTMAIN('CamCalib2','IRTCALIB_G1') ;
11:L P[1:ORIGIN] 100mm/sec FINE ;

```

D.9 Remove Calibration Grid Paddle

The application user frame teaching and camera calibration setup have been completed. Remove the grid paddle.

```

53: ;
54: !13)Remove grid from torch neck ;
55: ;

```

Make sure that the SETUP WIRE menu has correctly set up the number of wires for the torch. Default is a single wire torch.

D.10 Train User Tools

D.10.11 Train Master Single Wire User Tool

IRTSETUP_G1

```

79: ;
80: !SET MASTERS TO SINGLE ;
81: CALL IRTUTILITY_G1(140) ;

```

This command copies the Single wire user tool to the master single wire user tool. This gives the user a starting point for the master user tool.

IRTUTILITY_G1

```

208: LBL[140:SET UTOOL8 TO 4] ;
209: CALL IRTMATH('MovePR',0,0,(-77),77) ;
210: CALL IRTMAIN('SaveTool','Single',0,77) ;
211: CALL IRTMAIN('SetTool','MasterS',0,77) ;
212: CALL IRTMATH('MovePR',0,0,77,(-77)) ;
213: END ;

```

IRTSETUP_G1

The user can now manually override the WPR of the master tool. The current value is a coarse value.

For example if the Roll value is 93.45 it should probably be set to 90. If yaw is 154.34, it should probably be 150 assuming the torch is mounted at 30 degrees.

```

82:  ;
83:  !MANUALLY OVERRIDE MASTER ;
84:  ;
85:  !TEACH VIEW WITH 1 POINT ;
86:  CALL IRTUTILITY_G1(93) ;

```

D.10.12 Train View User Tool (for single wire)

IRTUTILITY_G1

The next step is to define the View Tool. (By default this is UTOOL 3). The view tool is used when moving the tool in the box. Its location is identical to the Single tool. However its orientation is not. Train this position as P[4].

IRTUTILITY_G1

If the robot has not been jogged it should already be in the correct position

```

126: LBL[93:SET VIEW] ;
127: CALL IRTMATH('MovePR',0,0,(-77),77) ;
128: CALL IRTMAIN('SetFrame','App') ;
129: CALL IRTMAIN('ToolZero') ;
130: PAUSE ;
131: !Train Wire with LIVE cross hairs ;
132:L P[4] 100mm/sec CNT100 ;
133: PR[77]=LPOS ;
134: CALL IRTMATH('Inverse',0,0,77,77) ;
135: CALL IRTMAIN('SetTool','View',0,77) ;
136: CALL IRTMATH('MovePR',0,0,77,(-77)) ;
137: END ;
138: ;
139: CALL IRTMATH('MovePR',0,0,(-77),77) ;
140: CALL IRTMATH('MovePR',0,0,(-78),78) ;
141: UFRAME_NUM=1 ;
142: UTOOL_NUM=0 ;
143: PR[77]=LPOS ;
144: CALL IRTMATH('Inverse',0,0,77,77) ;
145: UTOOL[3]=PR[77] ;
146: CALL IRTMATH('MovePR',0,0,77,(-77)) ;
147: CALL IRTMATH('MovePR',0,0,78,(-78)) ;
148: END ;

```

D.10.13 Train View User Tool for Tandem Torch

```

105: !TEACH VIEW WITH 1 POINT ;
106: CALL IRTUTILITY_G1(93) ;

```

IRTUTILITY_G1

This was previously discussed. P[4] will be taught midpoint between the two wires

```

        126:  LBL[93:SET VIEW] ;
        ...
148:  END ;

```

D.10.14 Train Master Dual Wire 1 User Tool

IRTSETUP_G1

```

107:  ;
108:  !SET MASTERS TO DUAL ;
109:  ;
110:  CALL IRTUTILITY_G1(120) ;

```

IRTUTILITY_G1

```

222:  LBL[120:SET UTOOL9 TO 1]
223:  CALL IRTMATH('MovePR',0,0,(-77),77)
224:  CALL IRTMAIN('SaveTool','Dual1',0,77)
225:  CALL IRTMAIN('SetTool','MasterD1',0,77)
226:  CALL IRTMATH('MovePR',0,0,77,(-77))
227:  END

```

D.10.15 Train Master Dual Wire 2 User Tool

Next we set the two master UTOOLS to the trained DUAL UTOOLS.

IRTSETUP_G1

```

111:  CALL IRTUTILITY_G1(160) ;

```

IRTUTILITY_G1

```

236:  LBL[160:SET UTOOL10 TO 2] ;
237:  CALL IRTMATH('MovePR',0,0,(-77),77) ;
238:  CALL IRTMAIN('SaveTool','Dual2',0,77) ;
239:  CALL IRTMAIN('SetTool','MasterD2',0,77) ;
240:  CALL IRTMATH('MovePR',0,0,77,(-77)) ;
241:  END ;

```

Manually override the WPR to values near the expected location

IRTSETUP_G1

```

112:  ;
113:  !MANUALLY OVERRIDE MASTER ;
114:  ;

```

D.11 Train Vision Process

```

115:  LBL[1030] ;
116:  ;
117:  !16)Jog Robot to have Wire tip ;
118:  !in the center of LIVE crosshair ;
119:  ;
120:  PAUSE ;
121:  !17)Train IRTWIRE_G1 Vision Proc ;

```

```

122: !inside IRT_G1 ;
123: CALL IRT_G1 ;

```

Call IRT_G1 to fine tune the UTOOLS

IRT_G1

```

1: CALL IRTMAIN('SetTorch',0,0,0,0,1) ;
2: ;

```

Define View Tool. The VIEW TOOL is set

```

3: CALL IRTUTILITY_G1(100) ;

```

There is nothing to modify in this program. It will update the view tool location based on the Wire tools.

IRTUTILITY_G1

```

150: LBL[100:UTOOL3 set ] ;
151: CALL IRTMAIN('INQUIRE', 'NTOOLS', 0, 0, 0, 77)
152: IF R[77]=2, JMP LBL[1002] ;
153: IF R[77]=1, JMP LBL[1001] ;
154: PAUSE ;
155: ABORT ;
156: ;

```

The View Tool (by default UTOOL[3] is set so the location is updated to match the SINGLE user tool.

```

157: LBL[1001:View = Single ] ;
158: CALL IRTMATH('MovePR',0,0,(-77),77) ;
159: CALL IRTMATH('MovePR',0,0,(-78),78) ;
160: CALL IRTMAIN('SetFrame', 'App') ;
161: CALL IRTMAIN('SaveTool', 'Single', 0, 77) ;
162: CALL IRTMAIN('SaveTool', 'View', 0, 78) ;
163: PR[78,1]=PR[77,1] ;
164: PR[78,2]=PR[77,2] ;
165: PR[78,3]=PR[77,3] ;
166: CALL IRTMAIN('SetTool', 'View', 0, 78) ;
167: CALL IRTMATH('MovePR',0,0,77,(-77)) ;
168: CALL IRTMATH('MovePR',0,0,78,(-78)) ;
169: END ;
170: ;

```

The View Tool (by default UTOOL[3] is set so the location is updated to match the midpoint of the two DUAL tools.

```

171: LBL[1002:Dual wire] ;
172: !Set UTOOL[3] to be midpoint of ;
173: !UTOOL[1] and UTOOL [2] ;

```

```

174: !Orientation from 1st tool ;
175: CALL IRTMATH('MovePR',0,0,(-77),77) ;
176: CALL IRTMATH('MovePR',0,0,(-78),78) ;
177: CALL IRTMATH('MovePR',0,0,(-79),79) ;
178: CALL IRTMATH('MovePR',0,0,(-80),80) ;
179: CALL IRTMAIN('SetFrame','App') ;
180: CALL IRTMAIN('SaveTool','Dual1',0,77) ;
181: CALL IRTMAIN('SaveTool','Dual2',0,78) ;
182: CALL IRTMAIN('SaveTool','View',0,79) ;
183: PR[80]=PR[77] ;
184: PR[80,1]=PR[77,1]+PR[78,1] ;
185: PR[80,1]=PR[80,1]/2 ;
186: PR[80,2]=PR[77,2]+PR[78,2] ;
187: PR[80,2]=PR[80,2]/2 ;
188: PR[80,3]=PR[77,3]+PR[78,3] ;
189: PR[80,3]=PR[80,3]/2 ;
190: CALL IRTMATH('Clear',0,0,77) ;
191: PR[79,1]=PR[80,1] ;
192: PR[79,2]=PR[80,2] ;
193: PR[79,3]=PR[80,3] ;
194: CALL IRTMAIN('SetTool','View',0,79) ;
195: CALL IRTMATH('MovePR',0,0,77,(-77)) ;
196: CALL IRTMATH('MovePR',0,0,78,(-78)) ;
197: CALL IRTMATH('MovePR',0,0,79,(-79)) ;
198: CALL IRTMATH('MovePR',0,0,80,(-80)) ;
199: END ;

```

Control returns to IRT_G1.

IRT_G1

```

4: LBL[10:Loop] ;
5: ;
6: !INSPECTION FRAME ;
7: CALL IRTMAIN('SetFrame','App') ;
8: !View UTOOL ;
9: CALL IRTMAIN('SetTool','View') ;

```

J[1] is the entrance/exit position of box, P[2] is view 1 position, and P[3] is view 2 position. None of these positions should need to be adjusted. At

```

10: J P[1:OVER BOX] 100% FINE ;
11: L P[2:VIEW -45 DEG] 200mm/sec FINE ;
12: !Run Snap View 1 R[1] is status ;

```

RUNSNAP1 finds the points of the wire at the view 1 position. Watch the runtime screen to verify that the wire is in the search window. Adjust the search window and repeat Line 13 if wire is not in search window.

```

13: CALL IRTMAIN('RUNSNAP1',0,1) ;
14: IF R[1]<>0, JMP LBL[100] ;

```

```
15:L P[3:VIEW 45 DEG] 2000mm/sec FINE ;
```

RUNSNAP2 finds the points of the wire at the view 2 position. Watch the runtime screen to verify that the wire is in the search window. Adjust the search window and repeat from Line 11 if wire is not in search window. (This will repeat RUNSNAP1 also).

```
16: !Run Snap View 2 R[1] is status ;
17: CALL IRTMAIN('RUNSNAP2',0,1) ;
18: IF R[1]<>0,JMP LBL[100] ;
```

The robot exits the box to P[1] and calls the GETTOOL function which defines the UTOOL(s). R[1] in Line 21 is defined as the status register.

```
19:L P[1:OVER BOX] 500mm/sec FINE ;
20: !GET TOOL R[1] is status ;
```

GETTOOL function calculates the User Tool

```
21: CALL IRTMAIN('GETTOOL',0,1) ;
```

Jump to different labels for each outcome

```
22: LBL[100] ;
23: SELECT R[1]=0,JMP LBL[500] ;
24:     =1,JMP LBL[501] ;
25:     =2,JMP LBL[502] ;
26:     =3,JMP LBL[503] ;
27:     =4,JMP LBL[504] ;
28:     =5,JMP LBL[505] ;
29:     ELSE,JMP LBL[510] ;
30: ;
31: LBL[500:Success] ;
32: LBL[502:Ignore] ;
33:L P[1:OVER BOX] 500mm/sec FINE ;
34: END ;
35: ;
36: LBL[503:Retry] ;
37: LBL[505:Retest] ;
38:L P[1:OVER BOX] 100mm/sec FINE ;
39: JMP LBL[10] ;
40: ;
41: LBL[501:Pending] ;
42: LBL[510:Invalid] ;
43:L P[1:OVER BOX] 500mm/sec FINE ;
44: ABORT ;
45: END ;
46: ;
47: LBL[504:Abort] ;
48:L P[1:OVER BOX] 500mm/sec FINE ;
49: ABORT ;
```

The UTOOLS have been updated SINGLE (UTOOL 4) or DUAL1 and DUAL2 (UTOOL 1 and UTOOL 2)

D.12 Update the Master Tools as Golden tools

IRTSETUP_G1

```
124:  ;
125:  PAUSE ;
126:  !19) Update Master tools ;
```

Determine if single or dual wire

```
127:  CALL IRTMAIN('INQUIRE', 'NTOOLS', 0, 0, 0, 77) ;
128:  IF R[77]=1, JMP LBL[2010] ;
129:  IF R[77]=2, JMP LBL[2020] ;
130:  PAUSE ;
131:  ABORT ;
132:  ;
```

Copy SINGLE value to MASTER. This should be the golden tool

```
133:  LBL[2010] ;
134:  !SET MASTERS TO SINGLE ;
135:  CALL IRTUTILITY_G1(140) ;
```

IRTUTILITY_G1

```
208:  LBL[140:SET UTOOL8 TO 4] ;
209:  CALL IRTMATH('MovePR', 0, 0, (-77), 77) ;
210:  CALL IRTMAIN('SaveTool', 'Single', 0, 77) ;
211:  CALL IRTMAIN('SetTool', 'MasterS', 0, 77) ;
212:  CALL IRTMATH('MovePR', 0, 0, 77, (-77)) ;
213:  END ;
```


Another opportunity to perfect the WPR of the MASTER tool.

IRTSETUP_G1

```
136:  ;
137:  !MANUALLY OVERRIDE MASTER ;
138:  JMP LBL[2030] ;
139:  ;
```

Set Master tool for UTOOL 1

```
140:  LBL[2020] ;
141:  !SET MASTERS TO DUAL ;
142:  ;143:  CALL IRTUTILITY_G1(120) ;
```

IRTUTILITY_G1

```
222:  LBL[120:SET UTOOL9 TO 1] ;
223:  CALL IRTMATH('MovePR',0,0,(-77),77) ;
224:  CALL IRTMAIN('SaveTool','Dual1',0,77) ;
225:  CALL IRTMAIN('SetTool','MasterD1',0,77) ;
226:  CALL IRTMATH('MovePR',0,0,77,(-77)) ;
227:  END ;
```

Set the Master tool for UTOOL 2.

IRTSETUP_G1

```
144:  CALL IRTUTILITY_G1(160) ;

236:  LBL[160:SET UTOOL10 TO 2] ;
237:  CALL IRTMATH('MovePR',0,0,(-77),77) ;
238:  CALL IRTMAIN('SaveTool','Dual2',0,77) ;
239:  CALL IRTMAIN('SetTool','MasterD2',0,77) ;
240:  CALL IRTMATH('MovePR',0,0,77,(-77)) ;
241:  END ;
```

Manually adjust the master tools WPR if desired.

IRTSETUP_G1

```
145:  ;
146:  !MANUALLY OVERRIDE MASTER ;
147:  ;
148:  LBL[2030] ;
149:  ;
150:  !Set Nominal Flag ;
151:  !now that tool is nominal ;
152:  !and re-run IRT_G1 ;
153:  CALL IRTUTILITY_G1(30) ;
```

D.13 Set the Nominal Tool (Golden Wire Tools)

Next we set the nominal flag before running IRT_G1 again

IRTUTILITY_G1

```
11: LBL[30:SET NOMINAL FLAG] ;
12: CALL IRTMAIN('SetNom',0,0,0,0,1) ;
13: END ;
```

IRTSETUP_G1

```
154: ;
155: CALL IRT_G1 ;
```

IRT G1 is executed again.

Should get the same results, except now the results are being saved as the nominal (Golden wire).

IRT_G1

```
1: CALL IRTMAIN('SetTorch',0,0,0,0,1) ;
2: ;
3: CALL IRTUTILITY_G1(100) ;
...
31: LBL[500:Success] ;

32: LBL[502:Ignore] ;
33: L P[1:OVER BOX] 500mm/sec FINE ;
34: END ;
```

IRTSETUP_G1

```
156: ;
157: !Nominal UT00L flag should be ;
158: !cleared automatically. ;
159: !From next inspection if UT ;
160: !Exceed thresholds, messages are ;
161: !posted in iRTorch Mate STATUS ;
162: ;
163: !----- SETUP COMPLETED ----- ;
```

You are done.

